

Mechanistic Interpretability of Analogical Reasoning in Gemma-2-2B: A Circuit Analysis

Abstract

We present a mechanistic interpretability analysis of analogical reasoning in Gemma-2-2B across 10 diverse A:B::C:? prompts. Attribution graphs were generated for each prompt via the Neuronpedia transcoder SAE (gemmascope-transcoder-16k), and 429 recurring circuit features were identified at a 50% appearance threshold. The 15 most universal features span layers 0–10, with the top features being: L6/F2586668 (structural/formal register, 48/10 graphs), L3/F5150441 (code-documentation patterns, 47/10), L8/F94882191 ("analogies or comparisons," 41/10), L4/F110446948 (license/code patterns, 37/10), and L7/F4828270 (reference codes and abbreviations, 31/10). Per-prompt causal path analysis reveals that model confidence correlates with path clarity: geographic capital-city analogies (Paris/Germany, $p=0.97$; Cairo/Kenya, $p=0.96$) show convergent 3–5-hop causal chains with strong domain-specific features, while functional/botanical analogies (Fish/air, $p=0.12$; Leaf/flower, $p=0.14$) produce diffuse, mixed-sign path structures with no clear dominant mechanism. Explicit analogy-detection features at Layers 7, 8, 9, and 15 are identified. Five of ten prompts contain purely excitatory causal paths; all ten show convergent late-layer aggregation through Layers 21–25 to the logit node at Layer 27. The results suggest that Gemma-2-2B implements analogical reasoning through a two-stage circuit: a universal structural-parsing stage (Layers 0–6) shared across all prompts, and a domain-specific resolution stage (Layers 7–27) that varies in strength with the clarity of the relational mapping.

Introduction

This paper presents a mechanistic interpretability analysis of analogical reasoning in Gemma-2-2B, a 2-billion parameter transformer language model. Analogical reasoning — the ability to identify and complete structural relationships of the form A:B::C:? — is a fundamental component of abstract intelligence. We investigate how this capability is implemented at the level of individual model features using attribution graphs generated via the Neuronpedia platform.

We selected 10 analogical reasoning prompts spanning diverse relational categories: geographic (capital cities), biological (animal/habitat, life stages), physical (tools/measurements), functional (activities), botanical (plant structures), mechanical (vehicle components), and social (institutional roles). For each prompt, we generated an attribution graph over Gemma-2-2B's transcoder features, then identified recurring circuit components across graphs using a 50% appearance threshold.

Our analysis reveals the specific layers and features that constitute the analogical reasoning circuit in Gemma-2-2B, the causal paths through which information flows from input tokens to the final predicted token, and the mechanistic basis for token selection in this class of reasoning task.

Methods

Model. Gemma-2-2B (Google DeepMind), a 26-layer transformer language model.

SAE. Gemmascope transcoder, 16k features per layer, applied at all transformer layers.

Graph generation. Attribution graphs were generated via the Neuronpedia API (`/api/graph/generate`) with parameters: `maxFeatureNodes=3000` , `desiredLogitProb=0.95` , `nodeThreshold=0.8` , `edgeThreshold=0.85` . Each graph captures the most influential transcoder features and their edge weights for a single prompt.

Recurring feature identification. `compare_graphs()` was run across all 10 graphs with `threshold=0.50` (a feature must appear in $\geq 5/10$ graphs to qualify as a shared circuit component). This threshold distinguishes category-level circuit structure from prompt-specific noise.

Node labeling. Recurring features were labeled via the Neuronpedia `/api/feature/{model}/{sae}/{feature}` endpoint, which returns human-readable descriptions derived from automated interpretability methods.

Causal path tracing. `trace_causal_paths()` traces the dominant signal pathways from influential input features to the logit node by following edge weights through the graph. Paths are scored by the product of absolute edge weights across hops and classified as excitatory (+), inhibitory (−), or mixed ($\pm$).

Per-prompt interpretation. `interpret_prompt_graph()` labels the top 40 nodes per graph (grouped into early/middle/late layer bands), identifies token competition, traces causal paths, generates a causal path figure, and appends all data to this paper.

Prompt dataset. 10 analogical reasoning prompts:

- 1. Paris is to France as Berlin is to
- 2. Cairo is to Egypt as Nairobi is to
- 3. Fish is to water as bird is to
- 4. Puppy is to dog as kitten is to
- 5. Clock is to time as thermometer is to
- 6. Book is to reading as radio is to
- 7. Leaf is to tree as petal is to
- 8. Wheel is to car as wing is to
- 9. Judge is to court as priest is to
- 10. Soldier is to army as sailor is to

Results

Recurring Circuit Features ($\geq 50\%$ of graphs, i.e., $\geq 5/10$)

A total of **429 recurring features** were identified at the 50% threshold. The top 15 by appearance count are shown below, with human-readable labels.

Layer	Feature	Node_id	Appearances	Avg Influence	Avg Activation	Label
6	2586668	6_2267_4	48/10	0.7217	15.7879	words that appear in programming code, legal jargon, or scientific texts
3	5150441	3_3205_3	47/10	0.6449	14.0235	code snippets and documentation references, possibly related to web development

8	94882191	8_13766_3	41/10	0.4939	32.1364	analogies or comparisons
4	110446948	4_14857_1	37/10	0.6755	15.9887	code snippets and license agreements
7	4828270	7_3099_1	31/10	0.7418	26.888	a variety of reference codes, abbreviations, and identifiers from different fields
9	89171325	9_13344_5	30/10	0.6755	13.7622	phrases suggesting uncertainty or comparison between two things
0	74438300	0_12200_1	27/10	0.701	3.8298	a variety of specific nouns
9	4261730	9_2909_8	27/10	0.6394	18.0839	formulas, ratios, and mathematical notation
5	16817094	5_5793_8	27/10	0.5668	17.3751	analogies
0	1708475	0_1847_1	26/10	0.5885	7.5277	scientific terms and experimental details related to biological and chemical research
0	81262125	0_12747_3	25/10	0.6496	3.8484	occurrences of the words 'they', 'them', 'that', 'these', or something that can be replaced with 'they' or 'them'
0	2239785	0_2115_4	25/10	0.637	6.5559	data reported as a percentage

						inside brackets, especially in a laboratory or medical context, and also recognizes countries
5	46836675	5_9672_7	25/10	0.5711	19.3411	the phrase "there/they are" or the phrase "it is to."
10	105902170	10_14542_8	23/10	0.7307	9.7973	words related to research, business or product comparison
7	286895	7_749_5	21/10	0.6201	12.1686	analogies and comparisons

Edge Neighbourhood Analysis (Step 5a)

Top feature by avg_influence: node_id=7_3099_1 (L7, F4828270, "reference codes, abbreviations, and identifiers")

Analysis performed on the graph for "Cairo is to Egypt as Nairobi is to".

- **In-degree:** 11, **Out-degree:** 7
- **Top incoming edges:** E_2_0 (embedding, $w=-38.73$), L4/F110446948 ($w=-13.61$), L3/F5150441 ($w=-6.45$), L6/F48733121 ($w=-5.42$), L5/F7993995 ($w=-4.95$), embedding/F1 ($w=+4.81$), L4/F47653198 ($w=+3.36$), L0/F26978184 ($w=+3.05$)
- **Top outgoing edges:** L10/F100614194 ($w=+1.84$), L9/F4701701 ($w=+1.22$), L10/F59891029 ($w=+1.10$), L9/F125286525 ($w=-1.01$), L18/F63861932 ($w=-0.22$), L27/F28273 ($w=-0.02$)

Key observations: This feature is strongly modulated by embedding-layer inputs (E_2_0 with weight -38.73), suggesting it fires in response to the specific token embeddings encoding the relation markers ("is to", "as"). Its strongest outgoing connections reach layers 10, 9, and ultimately layer 27, suggesting it participates in a long-range excitatory pathway toward the logit.

Second feature by avg_influence: node_id=6_2267_4 (L6, F2586668, "programming code, legal jargon, scientific texts")

Analysis performed on the graph for "Paris is to France as Berlin is to".

- **In-degree:** 7, **Out-degree:** 16
- **Top incoming edges:** embedding E_2_0 ($w=-7.33$), E_29437_1 ($w=-6.37$), E_6081_4 ($w=+3.92$), L5/F-1 ($w=+2.66$), E_603_2 ($w=+2.60$), L4/F110446948 ($w=-1.39$), L3/F-1 ($w=-1.08$)
- **Top outgoing edges:** L8/F45300912 ($w=+0.74$), L10/F133489619 ($w=+0.60$), L15/F55825445 ($w=+0.56$), L10/F19521865 ($w=+0.52$), L9/F63095751 ($w=+0.46$), L12/F2951222 ($w=+0.38$),

L8/F4947076 (w=+0.36), L9/F14810393 (w=+0.25)

Key observations: This feature fans out broadly to 16 downstream nodes across layers 8–18, acting as a hub that distributes structural/formal-register signal across many subsequent layers. It too is driven primarily by embedding-layer inputs. Its connection to the co-occurring recurring feature at L4/F110446948 (incoming, weight -1.39) reveals cross-inhibition between structural-register features at different layers.

Layer Analysis (Step 5b)

Hub layer: Layer 0 (appears 4 times in top-15 recurring features: F74438300, F1708475, F81262125, F2239785)

Layer 0 in the paris graph contains **146 nodes**. The top 15 by influence:

node_id	feature	influence	activation
0_3636_4	6615702	0.7993	2.1165
0_9222_4	42536475	0.7941	2.5155
0_8008_2	32076044	0.7937	1.3879
0_253_1	32384	0.7921	1.729
0_11834_8	70039529	0.7917	4.1844
0_12200_5	74438300	0.7915	2.5868
0_1548_5	1200474	0.7909	2.5233
0_0_7	-1	0.7903	0.0
0_5796_4	16805502	0.7892	2.0385
0_10210_5	52137365	0.7888	2.1873
0_4071_1	8292627	0.7859	1.4287
0_13919_4	96890159	0.7853	2.2555
0_3256_6	5305652	0.7834	2.7185
0_12698_1	80638649	0.7832	1.5973
0_9773_1	47770424	0.7829	1.8096

Layer 0's high density of recurring features reflects its role as a lexical-semantic gateway: at the first transformer layer, the model encodes noun identity, relational markers, and domain cues simultaneously. The recurring features at this layer (specific nouns, country recognition, pronoun resolution) suggest that the analogical circuit activates from the very first layer when it encounters the characteristic "X is to Y as Z is to" template.

Step 5c: Cross-graph Interpretation

Which features appear most universally? Features at layers 6 (F2586668, 48/10) and 3 (F5150441, 47/10) appear in nearly every graph. Interestingly, their labels ("programming code/legal jargon" and "code snippets/documentation") suggest these features detect the formal, structured register of the analogy

template rather than its semantic content. The "A is to B as C is to" construction has a syntactic formality that resembles technical documentation patterns.

Functional layer clustering:

- **Early layers (0–4):** Lexical encoding. Layer 0 fires broadly across token identities (nouns, pronouns, country names). Layers 3–4 detect structural/formal-register patterns in the prompt template.
- **Middle layers (5–9):** Relational abstraction. Layers 5, 7, 8, 9 host the explicit "analogy"/"comparison" features (F16817094, F286895, F94882191, F89171325) — the model is building an explicit representation that this is an analogy task.
- **Late layers (10+):** Answer convergence. Layer 10 (F105902170: "research, business or product comparison") and layer 25 participate in the final answer selection.

Surprising findings: The most universal features (L6/L3) are labeled with code/documentation patterns rather than analogy-specific labels, suggesting Gemma-2-2B's internal representation of analogy structure resembles formal-structured text patterns. The explicit "analogies" features (L5, L7, L8) are more semantically specific but appear later in the early/middle layers — the model first parses structure (L3–L6), then recognizes relational content (L7–L9).

Edge neighbourhood insights: The top feature by influence (7_3099_1) receives massive inhibitory input from embedding layers (–38.73), indicating strong competition between the embedding-level representation of the analogy template and this relational feature. Its outgoing path to layers 9–10 represents a compressed excitatory channel toward the answer.

Per-prompt Circuit Interpretations

The following sections are generated incrementally as each prompt is interpreted.

Prompt: "Paris is to France as Berlin is to"

Predicted token: Output " Germany" (p=0.973) (prob=0.9732)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
4	5757	4_5757_6	0.7999	2.0721	place names or location references
0	3636	0_3636_4	0.7993	2.1165	proper nouns, especially names of people and places
6	8956	6_8956_6	0.7991	4.4822	brand names, languages, and operating systems
4	15859	4_15859_6	0.7981	3.2247	places, people, and political entities in Europe and Africa, especially

					related to Swaziland, Denmark, and the Schleswig-Holstein region.
1	12602	1_12602_4	0.7975	2.9644	words associated with team sports, law, government, and time duration.
6	4516	6_4516_1	0.7973	7.8828	the letter "i."
3	3554	3_3554_6	0.7971	2.9865	mentions of countries
3	752	3_752_5	0.7969	2.3824	something about saving a file, or about volleyball, or about organizations with the initials WO.
1	6024	1_6024_6	0.7965	2.0856	words related to medicine & programming and 'ner' in various contexts
1	9357	1_9357_6	0.7963	2.9016	words or phrases related to booking flights, including origin, destination, airport, city, flight id and images
6	3335	6_3335_5	0.7957	4.8645	words and phrases that express difficulty or challenges
6	3182	6_3182_4	0.7953	3.475	mentions of cities or regions
4	15534	4_15534_6	0.7949	4.003	a variety of specific technical and scientific terms and jargon.
6	5451	6_5451_5	0.7947	5.2873	opinionated political discussions
4	13375	4_13375_5	0.7943	2.754	words and phrases that evoke a sense of nation and political entities
0	9222	0_9222_4	0.7941	2.5155	the equals sign
0	8008	0_8008_2	0.7937	1.3879	mentions of a female subject
4	426	4_426_5	0.7935	3.3206	a mix of positive and negative sentiment, and words related to quantity
5	3635	5_3635_6	0.7933	3.7195	proper nouns, especially names of places, people, languages, or organizations
5	4967	5_4967_8	0.7931	5.9053	common phrases with "out" and "back" along with some words indicating judgement.
5	5332	5_5332_4	0.7927	3.0736	proper nouns referring to geographic locations, organizations, and people.
0	2	0_2_7	0.7925	0.0	terminology related to technical and scientific writing

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
7	1022	7_1022_4	0.7997	4.2357	words ending with "ics" or "oma" as well as mentions of wicca.
10	5144	10_5144_5	0.7995	8.52	comparisons.
8	10532	8_10532_6	0.7989	8.4559	words referring to fields of study, types of problems in those fields, or people who work in those fields
13	4435	13_4435_8	0.7987	9.2783	opera-related terms, including German words and names of places
7	6884	7_6884_2	0.7979	9.3975	place names relating to France and cities, specifically "Saint-...", and "ville" or "canton."
7	11973	7_11973_8	0.7977	7.2224	mentions of tech companies and products, especially in the mobile device market
7	10892	7_10892_6	0.7967	5.0899	words related to babies, painful situations, or the body and/or reactions of the body.
7	2033	7_2033_6	0.7959	2.7182	mentions of legal trouble in Europe
7	3828	7_3828_5	0.7955	3.3997	scientific or technical language discussing experiments and programming; also finds celebrity names.
9	7669	9_7669_8	0.7945	6.4386	phrases that express strong disagreement or contradiction.
12	2171	12_2171_8	0.7939	5.9301	contexts where unit measurements are being discussed

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
19	14348	19_14348_8	0.8001	15.7641	uses of the word "to" and other prepositions, but not very precisely.
21	4827	21_4827_7	0.7985	45.0356	words and phrases related to scientific and medical research
15	8095	15_8095_6	0.7983	12.5808	people saying things, referencing political figures, locations, or general statements

16	6491	16_6491_8	0.7961	8.1436	locations, directions and distances, especially in California and Colorado.
0	17	0_17_7	0.7951	0.0	Spanish and Portuguese words related to code and computers
21	3271	21_3271_8	0.7929	7.8615	text about politics, history, and people
16	4240	16_4240_8	0.7923	8.8411	mentions of the United States and its economic standing in the world

Causal paths (edge-weight chains to logit):

*Path 1 — (±) mixed | weight=306.121568 | hops=3 → node_id=21_4827_7 | feature=4827 | Layer 21 | words and phrases related to scientific and medical research — edge [+197.9638] → node_id=22_15670_7 | feature=? | Layer 22 | (no label) — edge [-1.3555] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.1408] → **LOGIT** node_id=27_9066_8 Layer 27 — Output " Germany" (p=0.973)*

*Path 2 — (±) mixed | weight=178.849321 | hops=5 → node_id=16_6491_8 | feature=6491 | Layer 16 | locations, directions and distances, especially in California and Colorado. — edge [-0.7446] → node_id=17_14546_8 | feature=? | Layer 17 | (no label) — edge [+4.5501] → node_id=19_5773_8 | feature=? | Layer 19 | (no label) — edge [+2.9736] → node_id=21_7482_8 | feature=? | Layer 21 | (no label) — edge [+8.4876] → node_id=25_2725_8 | feature=? | Layer 25 | (no label) — edge [-2.0915] → **LOGIT** node_id=27_9066_8 Layer 27 — Output " Germany" (p=0.973)*

*Path 3 — (±) mixed | weight=168.024308 | hops=5 → node_id=7_11973_8 | feature=11973 | Layer 7 | mentions of tech companies and products, especially in the mobile device market — edge [+0.6996] → node_id=17_14546_8 | feature=? | Layer 17 | (no label) — edge [+4.5501] → node_id=19_5773_8 | feature=? | Layer 19 | (no label) — edge [+2.9736] → node_id=21_7482_8 | feature=? | Layer 21 | (no label) — edge [+8.4876] → node_id=25_2725_8 | feature=? | Layer 25 | (no label) — edge [-2.0915] → **LOGIT** node_id=27_9066_8 Layer 27 — Output " Germany" (p=0.973)*

*Path 4 — (±) mixed | weight=110.585287 | hops=4 → node_id=16_4240_8 | feature=4240 | Layer 16 | mentions of the United States and its economic standing in the world — edge [-2.0949] → node_id=19_5773_8 | feature=? | Layer 19 | (no label) — edge [+2.9736] → node_id=21_7482_8 | feature=? | Layer 21 | (no label) — edge [+8.4876] → node_id=25_2725_8 | feature=? | Layer 25 | (no label) — edge [-2.0915] → **LOGIT** node_id=27_9066_8 Layer 27 — Output " Germany" (p=0.973)*

*Path 5 — (±) mixed | weight=65.927255 | hops=4 → node_id=6_3335_5 | feature=3335 | Layer 6 | words and phrases that express difficulty or challenges — edge [+0.2497] → node_id=19_855_8 | feature=? | Layer 19 | (no label) — edge [+13.5011] → node_id=24_1963_8 | feature=? | Layer 24 | (no label) — edge [-9.3496] → node_id=25_2725_8 | feature=? | Layer 25 | (no label) — edge [-2.0915] → **LOGIT** node_id=27_9066_8 Layer 27 — Output " Germany" (p=0.973)*

Causal path diagram:

 Causal paths for "<bos>Paris is to France as Berlin is to" → "Output " Germany" (p=0.973)"

Figure: Causal paths from input features to predicted token "Output " Germany" (p=0.973)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] At the earliest layers, Gemma-2-2B immediately parsed the geographic and political character of the input. Feature F5757 at Layer 4 (node_id=4_5757_6, inf=0.7999) registered "place names or location references," while F15859 at Layer 4 (node_id=4_15859_6, inf=0.7981) more specifically detected "places, people, and political entities in Europe and Africa." Feature F3554 at Layer 3 (node_id=3_3554_6, inf=0.7971) activated for "mentions of countries," and F13375 at Layer 4 (node_id=4_13375_5, inf=0.7943) for "words and phrases that evoke a sense of nation and political entities." Together, these early features establish that the model has identified all four terms (Paris, France, Berlin) as European geographic and political entities before any relational processing begins. Notably, F3636 at Layer 0 (node_id=0_3636_4, inf=0.7993) fires for "proper nouns, especially names of people and places" — the very first transformer layer is already registering proper-noun status for all analogy terms.

[MIDDLE] By the middle layers, domain knowledge takes over with two striking developments. First, F6884 at Layer 7 (node_id=7_6884_2, inf=0.7979) activates specifically for "place names relating to France and cities, specifically 'Saint-...' and 'ville' or 'canton'" — the model has mapped the A-term (Paris) into its France-specific geography domain. Then, F4435 at Layer 13 (node_id=13_4435_8, inf=0.7987) fires for "opera-related terms, including German words and names of places," grounding the B-term (Berlin) in its German cultural context. The critical relational insight arrives at L10 with F5144 (node_id=10_5144_5, inf=0.7995), labeled simply "comparisons" — by Layer 10, the model has abstracted the analogy template as a comparison structure, with Paris-to-France and Berlin-to-Germany resolved as parallel capital-to-country relationships.

[LATE] The late layers crystallise the answer through a convergent pathway. F14348 at Layer 19 (node_id=19_14348_8, inf=0.8001) — labeled "uses of the word 'to' and other prepositions" — achieves the highest influence score in the entire graph. This feature appears to be tracking the structural connector "is to" in the analogy template, acting as a final syntactic anchor. F4827 at Layer 21 (node_id=21_4827_7, inf=0.7985) provides the dominant upstream push to the logit via Path 1 (weight=306.1), sending an edge of +197.96 through Layer 22 to Layer 25, which then fires the logit node with +1.14.

[TOKEN COMPETITION] The logit was predicted with probability 0.9732 — an extraordinarily confident prediction, reflecting the clarity of the capital-city relationship. The circuit was convergent: Paths 2–4 all share the L17→L19→L21→L25→LOGIT backbone, demonstrating that multiple early-to-mid layer triggers (a location feature at L16/F6491, a U.S.-economy feature at L16/F4240, and tech-company features at L7/F11973) all route their signals through the same late-layer aggregation channels. All five paths are classified as Mixed (alternating positive and negative edges), but the dominant direction is excitatory — the negative edges (at L22→L25 and L25→LOGIT transitions in Path 1) represent competition suppression rather than true inhibition of the target token.

[SYNTHESIS] The complete causal story: the model detects European geographic entities at Layer 0–4, maps France-specific and Germany-specific geographic knowledge at Layers 7–13, recognises the comparison template at Layer 10, and routes converging signals through a shared L17→L21→L25 aggregation pathway to the logit. The dominant mechanism is Path 1, originating at the "scientific/medical research" feature at Layer 21 (F4827), which despite its seemingly mismatched label, carries the strongest single edge weight (+197.96) to the logit. This suggests the model encodes the formal, structured nature of the analogy template in a feature cluster associated with technical register. The extraordinary confidence

(p=0.9732) reflects a clean convergent circuit with no genuine competitor paths — "Germany" was the unambiguous winner.

Prompt: "Cairo is to Egypt as Nairobi is to"

Predicted token: Output " Kenya" (p=0.963) (prob=0.9628)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
5	7130	5_7130_5	0.7996	4.603	verbs ending in "ing" that are being used as gerunds or present participles
3	15381	3_15381_2	0.7994	3.3469	names of government departments and related terms
0	1	0_1_7	0.7991	0.0	Spanish-language programming-related terms
5	15362	5_15362_5	0.7989	2.5296	mentions of people and places
4	9704	4_9704_4	0.7987	3.2569	words that describe environmental conditions or physical attributes, and also some measurement or lab related terms
1	9900	1_9900_6	0.7981	2.2453	words related to booking or reserving something
5	14698	5_14698_6	0.7979	2.7607	places, especially universities and US states
5	575	5_575_5	0.7973	2.9661	code snippets or descriptions of converting digital signals to analog
5	6490	5_6490_1	0.7971	4.0009	words and phrases related to geography, politics, or groups of people, particularly related to conflict
4	12832	4_12832_5	0.7969	2.3034	words or short phrases in the neighborhood of "she," "he," "I," "as".

0	0	0_0_7	0.7965	0.0	mentions of clubs or sports teams, and sometimes related words like 'sister' or 'kids'
2	13965	2_13965_1	0.7963	2.6666	places on a map, body parts involved in medicine, and economic terms
5	11012	5_11012_3	0.7954	5.2732	verbs of action and change
5	6728	5_6728_5	0.7952	3.7843	code, RDF, and scientific terms
0	5215	0_5215_4	0.795	2.1942	code-related terms, especially from multiple languages (C, Swift, etc.) and technical documentation.
1	7832	1_7832_3	0.7948	1.5438	the word "to".
0	4470	0_4470_4	0.7946	1.9006	references to religion, religious figures, or important religious places
3	11806	3_11806_5	0.7936	2.2821	words/phrases related to story telling
5	5500	5_5500_8	0.7934	3.6353	profanity and comparisons
0	14184	0_14184_5	0.7926	3.372	the word "it" when immediately preceded by a word ending in a vowel and immediately followed by a word starting with a consonant
0	13916	0_13916_6	0.7924	2.8652	words and terms about medicine and scientific research, including references.

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
10	14118	10_14118_8	0.8	5.8252	phrases expressing unfavorable conditions or uncertainty
8	15706	8_15706_6	0.7998	3.9033	words related to markets, measurements, and racial issues
6	2736	6_2736_8	0.7993	7.8579	words associated with negativity in society, relationships and situations.
11	3596	11_3596_4	0.7983	9.5114	snippets of code and text relating to academic research
7	11758	7_11758_4	0.7967	5.0055	words with some cultural (often middle eastern) or historical aspect that may be a name or location

0	8	0_8_7	0.7959	0.0	phrases with "are," and sometimes also finds other words related to research, science, testing, and data
6	10428	6_10428_6	0.7956	2.8565	references to animals or young humans
7	1177	7_1177_6	0.7944	3.4789	phrases related to US politics, history, and terrorism
6	9865	6_9865_1	0.794	13.5981	code and file paths
0	9	0_9_1	0.7932	0.0	math and mathematical notation
6	3192	6_3192_1	0.793	5.7159	proper nouns that are countries or cities

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
18	12171	18_12171_6	0.7985	16.2289	language that indicates unfairness or a lack of fairness
15	5355	15_5355_4	0.7977	10.3601	geographical locations
0	15	0_15_2	0.7975	0.0	the word "interval" and phrases containing it
13	10167	13_10167_8	0.7961	6.7758	words relating to student life, sports, experiments and medicine
13	4435	13_4435_8	0.7958	11.0511	opera-related terms, including German words and names of places
18	13586	18_13586_8	0.7942	15.5628	Legal documents
14	8796	14_8796_4	0.7938	25.0068	places, countries, areas, and tribes, particularly in the middle east
15	1019	15_1019_6	0.7928	14.5685	sentences with a lot of clauses, logic and/or comparisons.

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=86.124859 / hops=5 → node_id=13_4435_8 | feature=4435 | Layer 13 | opera-related terms, including German words and names of places — edge [+2.2185] → node_id=15_15954_8 | feature=? | Layer 15 | (no label) — edge [+6.7443] → node_id=18_13586_8 | feature=13586 | Layer 18 | Legal documents — edge [+1.5251] → node_id=24_1260_8 | feature=? | Layer 24 | (no label) — edge [-8.5025] → node_id=25_15920_8 | feature=? | Layer 25 | (no label) — edge [-0.4439] → **LOGIT** node_id=27_28273_8 Layer 27 — Output " Kenya" (p=0.963)

Path 2 — (±) mixed / weight=67.481537 / hops=5 → node_id=5_15362_5 | feature=15362 | Layer 5 | mentions of people and places — edge [+0.1594] → node_id=20_15360_8 | feature=? | Layer 20

| (no label) — edge [+7.7863] → node_id=23_13914_8 | feature=? | Layer 23 | (no label) — edge [+10.6511] → node_id=24_8251_8 | feature=? | Layer 24 | (no label) — edge [-4.762] → node_id=25_286_8 | feature=? | Layer 25 | (no label) — edge [-1.0719] → **LOGIT**
node_id=27_28273_8 Layer 27 — Output " Kenya" (p=0.963)

Path 3 — (±) mixed / weight=60.1826 / hops=4 → node_id=0_8_7 | feature=8 | Layer 0 | phrases with "are," and sometimes also finds other words related to research, science, testing, and data — edge [+1.9875] → node_id=10_5559_8 | feature=? | Layer 10 | (no label) — edge [-22.7107] → node_id=11_16076_8 | feature=? | Layer 11 | (no label) — edge [+3.0036] → node_id=25_15920_8 | feature=? | Layer 25 | (no label) — edge [-0.4439] → **LOGIT**
node_id=27_28273_8 Layer 27 — Output " Kenya" (p=0.963)

Path 4 — (±) mixed / weight=36.433288 / hops=5 → node_id=5_5500_8 | feature=5500 | Layer 5 | profanity and comparisons — edge [+0.9529] → node_id=7_749_8 | feature=? | Layer 7 | (no label) — edge [+1.2626] → node_id=10_5559_8 | feature=? | Layer 10 | (no label) — edge [-22.7107] → node_id=11_16076_8 | feature=? | Layer 11 | (no label) — edge [+3.0036] → node_id=25_15920_8 | feature=? | Layer 25 | (no label) — edge [-0.4439] → **LOGIT**
node_id=27_28273_8 Layer 27 — Output " Kenya" (p=0.963)

Path 5 — (+) excitatory / weight=35.325629 / hops=4 → node_id=15_5355_4 | feature=5355 | Layer 15 | geographical locations — edge [+0.7159] → node_id=20_5236_8 | feature=? | Layer 20 | (no label) — edge [+3.5751] → node_id=21_5251_8 | feature=? | Layer 21 | (no label) — edge [+9.6702] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.4273] → **LOGIT**
node_id=27_28273_8 Layer 27 — Output " Kenya" (p=0.963)

Causal path diagram:

 Causal paths for "<bos>Cairo is to Egypt as Nairobi is to" → "Output " Kenya" (p=0.963)"

Figure: Causal paths from input features to predicted token "Output " Kenya" (p=0.963)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] For "Cairo is to Egypt as Nairobi is to," the early layers encode a more diffuse set of features than for the Paris/Germany analogy. The highest-influence early node is F7130 at Layer 5 (node_id=5_7130_5, inf=0.7996), detecting "verbs ending in 'ing' used as gerunds or present participles" — a structural parsing feature responding to the "is to" relational connectors. F15381 at Layer 3 (node_id=3_15381_2, inf=0.7994) activates for "names of government departments and related terms," reflecting the model's encoding of the political-administrative relationship between capitals and countries. F15362 at Layer 5 (node_id=5_15362_5, inf=0.7989) catches "mentions of people and places," broadly registering the proper-noun geography. F14698 at Layer 5 (node_id=5_14698_6, inf=0.7979) specifically activates for "places, especially universities and US states," while F7 at Layer 7 (middle band) fires for "words with some cultural (often middle eastern) or historical aspect" — registering that Cairo and Egypt are Middle Eastern proper nouns.

[MIDDLE] The middle layers show a surprising pattern: the highest-influence middle node, F14118 at Layer 10 (node_id=10_14118_8, inf=0.8000), is labeled "phrases expressing unfavorable conditions or uncertainty," and F15706 at Layer 8 (node_id=8_15706_6, inf=0.7998) detects "words related to markets, measurements, and racial issues." These labels seem mismatched to a geographic analogy — but this misalignment may reflect that the model's recurring "comparison/uncertainty" features activate broadly for any A:B::C:? structure, since comparison inherently involves contrasting two things, which the model internally represents through a negative-framing feature. More directly geographic is F14796 at Layer 14 in the late

band: "places, countries, areas, and tribes, particularly in the middle east" (node_id=14_8796_4, inf=0.7938), which explicitly maps Nairobi into an East African geographic context.

[LATE] The late layers are dominated by F12171 at Layer 18 (node_id=18_12171_6, inf=0.7985) — "language that indicates unfairness or a lack of fairness" — another apparently mismatched label that likely reflects this feature activating on the contrastive-comparison structure. The key geographic feature is F5355 at Layer 15 (node_id=15_5355_4, inf=0.7977), labeled "geographical locations," which forms the origin of the only purely excitatory path (Path 5, weight=35.3): L15/F14426490 (+0.72) → L20 (+3.58) → L21 (+9.67) → L25 (+1.43) → LOGIT. This ascending-weight excitatory chain (0.72 → 3.58 → 9.67 → 1.43) shows signal being amplified through the late layers toward the predicted token " Kenya."

[TOKEN COMPETITION] The prediction " Kenya" was made with probability 0.963, indicating high confidence. Path 5 (excitatory, weight=35.3) was the only pure excitatory path among five identified paths; the remaining four were all Mixed. The dominant Path 1 (mixed, weight=86.1) traces from L13/F9899011 through layers 15, 18, 24, 25 to the logit — a longer convergent chain involving "opera-related terms, including German words and names of places" (L13/F4435) as an intermediate node, suggesting the model uses a general European/African proper-noun feature cluster to mediate geographic capital-to-country relationships.

[SYNTHESIS] The complete causal story for "Cairo is to Egypt as Nairobi is to" → " Kenya": the model detects the proper-noun geographic and administrative nature of the terms at Layers 3–5, activates Middle Eastern/African geographic features at Layers 7–14, then routes through a convergent late-layer aggregation (L17→L20→L21→L25→LOGIT) to produce the high-confidence answer. The dominant mechanism is Path 1 (weight=86.1, mixed), supplemented by the only clean excitatory path (Path 5, weight=35.3) originating from a "geographical locations" feature at Layer 15. The circuit is convergent, with multiple paths sharing the same L24→L25→LOGIT backbone, but relies more heavily on mixed-sign edge chains than the Germany analogy — consistent with the model's internal representation of African geographic relationships through indirect feature clusters rather than the direct Europe-specific features available for the Paris/Berlin prompt.

Prompt: "Fish is to water as bird is to"

Predicted token: Output " air" (p=0.117) (prob=0.1167)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
6	9865	6_9865_3	0.7997	20.1705	code and file paths
4	2866	4_2866_2	0.7989	7.1793	the word "is"
2	9088	2_9088_3	0.7985	1.6687	things which are believed, demonstrated, or thought to be true
2	426	2_426_1	0.7976	2.6844	words related to computer programming, math, or numbers

3	5102	3_5102_2	0.7974	3.7084	technical jargon and scientific terms from a wide variety of fields
0	14963	0_14963_6	0.7972	3.7276	words related to medical/legal/scientific terminology
2	7356	2_7356_4	0.7966	2.505	cooking ingredients and recipes
5	4703	5_4703_4	0.7962	4.7563	metaphors for dangerous or surprising events
1	8120	1_8120_6	0.796	3.8706	recipes and food preparation techniques
4	2857	4_2857_8	0.7959	4.439	verbs related to tasks and performing them
5	7191	5_7191_8	0.7957	5.7456	words and phrases related to legal proceedings or agreements
6	5621	6_5621_5	0.7955	3.57	descriptions of musical albums and sports leagues and seasons
1	11752	1_11752_4	0.7951	2.8279	scientific names of organisms that cause plant galls
0	8163	0_8163_6	0.7945	4.5913	words and phrases related to societal and political issues
0	8008	0_8008_2	0.7943	1.3318	mentions of a female subject
5	9672	5_9672_4	0.7939	3.5277	the phrase "there/they are" or the phrase "it is to."
4	9720	4_9720_3	0.7933	4.8912	announcements of product or company news and clinical trials
4	6466	4_6466_5	0.7931	3.1166	legal and medical terminology
6	12493	6_12493_5	0.7929	7.5923	legal citations
6	11763	6_11763_4	0.7928	3.5949	substrings resembling code or equations including special characters
5	11877	5_11877_5	0.7926	3.8685	mathematical terms related to exponents and powers

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
7	13919	7_13919_8	0.7995	8.2225	words related to regulations, standards, and procedures.

9	8857	9_8857_5	0.7993	6.6666	answer choices in mathematical questions
8	4440	8_4440_2	0.7991	12.7045	discussions of food, especially meat
7	2318	7_2318_8	0.7983	3.2918	a variety of numeric and statistical data, including p-values, BMI, ages, and measurements
8	13494	8_13494_6	0.7982	5.5575	phrases related to running out of supplies
7	2678	7_2678_8	0.7978	5.6884	a wide variety of mostly short phrases; the neuron may be faulty
7	7080	7_7080_4	0.797	4.5232	words and phrases related to food and dining, especially in the context of a region or place.
13	2249	13_2249_6	0.7964	13.5967	noun phrases ending in "-ists", and also seems to have some other unrelated triggers.
7	8622	7_8622_6	0.7949	5.8154	words related to animals, taxonomy, animal habitats and animal research
7	4108	7_4108_8	0.7937	5.6759	verbs and prepositions possibly related to conflict or legal disputes
7	11973	7_11973_8	0.7935	7.516	mentions of tech companies and products, especially in the mobile device market

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
0	22	0_22_5	0.8	0.0	words ending in the suffix "ality," or the word "other."
14	14321	14_14321_8	0.7999	6.926	words commonly found at the beginning of sentences or phrases
16	12036	16_12036_8	0.7987	7.1728	a little bit of everything, including punctuation, 's', and the words 'uniform' and 'after', which is too broad to be useful
17	101	17_101_8	0.798	9.8202	non-English words
21	881	21_881_8	0.7968	7.8511	connectives and technical terms
15	12472	15_12472_4	0.7953	60.6484	words associated with legal, academic, or technical documents, and numerical references

17	6986	17_6986_8	0.7947	9.8884	things wrapped or attached to other things, especially with the preposition "on."
0	16	0_16_3	0.7941	0.0	words related to political administration and legal rights

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=60.808751 / hops=2 → node_id=E_2003_4 | feature=? | Layer E | (no label) — edge [+12.0646] → node_id=22_4252_8 | feature=? | Layer 22 | (no label) — edge [+5.0403] → LOGIT node_id=27_2681_8 Layer 27 — Output " air" (p=0.117)

Path 2 — (±) mixed / weight=41.67803 / hops=4 → node_id=1_8120_6 | feature=8120 | Layer 1 | recipes and food preparation techniques — edge [+0.6877] → node_id=4_13402_6 | feature=? | Layer 4 | (no label) — edge [+15.4161] → node_id=20_3824_8 | feature=? | Layer 20 | (no label) — edge [+14.4137] → node_id=25_11801_8 | feature=? | Layer 25 | (no label) — edge [-0.2727] → LOGIT node_id=27_2681_8 Layer 27 — Output " air" (p=0.117)

Path 3 — (±) mixed / weight=22.713631 / hops=4 → node_id=2_7356_4 | feature=7356 | Layer 2 | cooking ingredients and recipes — edge [+0.2426] → node_id=22_4252_8 | feature=? | Layer 22 | (no label) — edge [+19.4543] → node_id=24_8106_8 | feature=? | Layer 24 | (no label) — edge [-6.0286] → node_id=25_12962_8 | feature=? | Layer 25 | (no label) — edge [-0.7984] → LOGIT node_id=27_2681_8 Layer 27 — Output " air" (p=0.117)

Path 4 — (±) mixed / weight=1.801989 / hops=3 → node_id=8_13494_6 | feature=13494 | Layer 8 | phrases related to running out of supplies — edge [+0.3744] → node_id=24_8106_8 | feature=? | Layer 24 | (no label) — edge [-6.0286] → node_id=25_12962_8 | feature=? | Layer 25 | (no label) — edge [-0.7984] → LOGIT node_id=27_2681_8 Layer 27 — Output " air" (p=0.117)

Causal path diagram:



Causal paths for "<bos>Fish is to water as bird is to" → "Output " air" (p=0.117)"

Figure: Causal paths from input features to predicted token "Output " air" (p=0.117)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] For "Fish is to water as bird is to," the early-layer features engage primarily with structural and syntactic parsing. The top early node, F9865 at Layer 6 (node_id=6_9865_3, inf=0.7997), is labeled "code and file paths" -- a structural-register feature that fires on the formal pattern of the analogy template. F2866 at Layer 4 (node_id=4_2866_2, inf=0.7989) detects "the word 'is'" -- the model is explicitly tracking the relational connector. F9088 at Layer 2 (node_id=2_9088_3, inf=0.7985) activates for "things which are believed, demonstrated, or thought to be true," reflecting the declarative, definitional nature of analogy frames. Crucially absent are any domain-specific features for fish, water, or birds -- the early circuit is operating almost entirely on the syntactic structure of the analogy.

[MIDDLE] The middle layers show no explicit habitat/environment features. F13919 at Layer 7 (node_id=7_13919_8, inf=0.7995) fires for "words related to regulations, standards, and procedures" -- another structural-formalism feature. F8857 at Layer 9 (node_id=9_8857_5, inf=0.7993) detects "answer

choices in mathematical questions," and F8494 at Layer 8 fires for "food, especially meat." These labels are unexpected for a habitat analogy -- they suggest the model's internal circuit for analogical reasoning uses features that generalize across many formal-comparison contexts rather than encoding domain-specific knowledge.

[LATE] The most interesting late-layer finding: F22 at Layer 0 (node_id=0_22_5, inf=0.8000) -- labeled "words ending in the suffix '-ality' or the word 'other'" -- achieves the highest influence score. This may reflect the model's encoding of the open-completion slot ("is to ____") as an abstract category placeholder. F14321 at Layer 14 (node_id=14_14321_8, inf=0.7999) fires for "words commonly found at the beginning of sentences or phrases," suggesting it is tracking the output-initiation context. F881 at Layer 21 (node_id=21_881_8, inf=0.7968) detects "connectives and technical terms."

[TOKEN COMPETITION] The model predicted " air" with only 11.7% probability -- the lowest confidence of all 10 prompts. The dominant excitatory path (Path 1, weight=60.8) is remarkably short: embedding -> L22/F9139927 -> LOGIT with edge weights 12.06 and 5.04. This 2-hop path bypasses all the labeled features entirely, suggesting that for this ambiguous analogy, the final token decision was driven by a direct embedding-to-late-layer route rather than the full multi-layer causal chain seen in high-confidence predictions.

[SYNTHESIS] This is the most ambiguous circuit in the dataset. The model has no strong domain-specific features for "air" or "sky" -- both are valid completions. The circuit is operating in a near-pure structural mode: the analogy template is recognized (structural features at L2-L6), the comparison slot is registered (F22 as abstract placeholder), and the logit is driven by a 2-hop excitatory path from an embedding feature. The 11.7% confidence reflects genuine competition: "sky," "air," and other habitat words have roughly equal support from the circuit, and the final choice of " air" appears to be driven by subtle token-level frequency biases rather than a strong domain-specific causal chain.

Prompt: "Puppy is to dog as kitten is to"

Predicted token: Output " cat" (p=0.756) (prob=0.7565)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	11438	1_11438_5	0.8	2.3273	the word "as", and in some contexts words related to government organization or proceedings
1	12148	1_12148_6	0.7997	1.9932	words related to hosts, pathogens, and infections
3	5267	3_5267_4	0.7995	2.8455	words related to science and engineering that are nouns
5	7714	5_7714_8	0.7991	5.2637	words that are used to evaluate a situation, including words indicating negativity, scientific discourse,

					mathematical language and general business reviews about internet companies
5	13340	5_13340_4	0.7989	4.9129	technical or jargonistic language related to a variety of fields
3	308	3_308_2	0.7983	2.4183	instances of writing, commenting, and communicating about technical subjects
4	11886	4_11886_5	0.7981	2.6228	conditional and definitional statements within mathematical text
1	1250	1_1250_5	0.7975	2.0992	LaTeX or XML markup
0	3256	0_3256_6	0.7965	3.7466	SSH public key authentication details
7	10892	7_10892_6	0.7959	4.3184	words related to babies, painful situations, or the body and/or reactions of the body.
7	10892	7_10892_8	0.7957	6.9252	words related to babies, painful situations, or the body and/or reactions of the body.
2	11571	2_11571_1	0.7955	2.3846	words related to family, childhood, and male relationships
6	9865	6_9865_3	0.7953	20.8974	code and file paths
4	11188	4_11188_3	0.7951	3.9347	the word "so" and sometimes the words "too" or "am" near "so."
7	2678	7_2678_8	0.7949	4.8255	a wide variety of mostly short phrases; the neuron may be faulty
7	462	7_462_1	0.7947	4.3856	code documentation and import statements
1	11387	1_11387_5	0.7943	2.3956	formal legal definitions and analysis of how principles work
2	14798	2_14798_2	0.7941	3.031	sentences using helping verbs, along with content related to criminal or political circumstances.
0	127	0_127_1	0.7939	1.4752	proper names, particularly of people and places.
7	3099	7_3099_6	0.7935	12.1392	a variety of reference codes, abbreviations, and identifiers from different fields.
3	15856	3_15856_8	0.7933	5.556	conjunctions (and/or/but) near words indicating possibility

0	11710	0_11710_6	0.7929	2.3997	words, abbreviations, and symbols related to scientific measurement and notation
6	2267	6_2267_2	0.7927	20.2967	words that appear in programming code, legal jargon, or scientific texts
5	4539	5_4539_4	0.7925	5.0999	words and phrases related to peer-to-peer networks, scientific study of hosts/pathogens, and physics
6	14666	6_14666_6	0.7923	3.643	terms relating to baseball and other sports teams, organizations, and competitions

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
9	2909	9_2909_7	0.7999	14.7521	formulas, ratios, and mathematical notation
10	2565	10_2565_4	0.7987	7.6054	commonly used sayings or idioms
14	4256	14_4256_6	0.7985	11.3468	a group of words; one of which is "astronomy", and the rest of which are from a list of words about art, measurement, time and history.
10	14542	10_14542_6	0.7979	8.8969	words related to research, business or product comparison
11	9508	11_9508_8	0.7977	9.5501	mathematical expressions with variables and subscripts especially coordinate pairs contained within parentheses.
11	5517	11_5517_4	0.7973	9.6624	a wide array of words in a text.
9	16195	9_16195_6	0.7971	5.4992	words related to food, cooking, and building materials
9	8770	9_8770_1	0.7967	23.1286	sentences beginning with "It" or "In" and also the word "There".
13	735	13_735_3	0.7963	13.3752	the word "to", and some other prepositions
10	8919	10_8919_6	0.7945	3.1049	vocabulary related to strong emotions and prejudice.
10	15902	10_15902_4	0.7937	5.6055	words describing organizations and their relationships to issues

9	1195	9_1195_8	0.7931	7.2028	various political or social topics, including scandals, film festivals, and finance
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Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
0	22	0_22_6	0.7993	0.0	words ending in the suffix "ality," or the word "other."
25	553	25_553_8	0.7969	32.9351	text passages written in all capital letters
23	6905	23_6905_8	0.7961	12.4826	nouns preceded by articles or adjectives within a specific context, often related to political or public discourse

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=551.738936 / hops=5 → node_id=9_2909_7 | feature=2909 | Layer 9 | formulas, ratios, and mathematical notation — edge [-1.2561] → node_id=14_3704_7 | feature=? | Layer 14 | (no label) — edge [+11.8871] → node_id=20_3094_7 | feature=? | Layer 20 | (no label) — edge [-12.7578] → node_id=22_15670_7 | feature=? | Layer 22 | (no label) — edge [-2.4502] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.1821] → **LOGIT** node_id=27_4401_8 Layer 27 — Output " cat" (p=0.756)

Path 2 — (+) excitatory / weight=294.013773 / hops=2 → node_id=E_56081_6 | feature=? | Layer E | (no label) — edge [+49.1547] → node_id=19_302_8 | feature=? | Layer 19 | (no label) — edge [+5.9814] → **LOGIT** node_id=27_4401_8 Layer 27 — Output " cat" (p=0.756)

Path 3 — (±) mixed / weight=194.912395 / hops=5 → node_id=13_735_3 | feature=735 | Layer 13 | the word "to", and some other prepositions — edge [+5.717] → node_id=17_13553_4 | feature=? | Layer 17 | (no label) — edge [+6.0761] → node_id=18_6875_4 | feature=? | Layer 18 | (no label) — edge [-4.0453] → node_id=21_15785_4 | feature=? | Layer 21 | (no label) — edge [+0.2942] → node_id=24_10034_8 | feature=? | Layer 24 | (no label) — edge [-4.7146] → **LOGIT** node_id=27_4401_8 Layer 27 — Output " cat" (p=0.756)

Path 4 — (±) mixed / weight=73.6242 / hops=5 → node_id=14_4256_6 | feature=4256 | Layer 14 | a group of words; one of which is "astronomy", and the rest of which are from a list of words about art, measurement, time and history. — edge [-0.9386] → node_id=17_12910_6 | feature=? | Layer 17 | (no label) — edge [+6.3577] → node_id=19_302_6 | feature=? | Layer 19 | (no label) — edge [+5.6374] → node_id=22_7486_8 | feature=? | Layer 22 | (no label) — edge [+8.9144] → node_id=25_15216_8 | feature=? | Layer 25 | (no label) — edge [-0.2455] → **LOGIT** node_id=27_4401_8 Layer 27 — Output " cat" (p=0.756)

Path 5 — (±) mixed / weight=57.234595 / hops=4 → node_id=10_8919_6 | feature=8919 | Layer 10 | vocabulary related to strong emotions and prejudice. — edge [+0.7907] → node_id=19_302_8 | feature=? | Layer 19 | (no label) — edge [+33.0763] → node_id=22_7486_8 | feature=? | Layer 22 | (no label) — edge [+8.9144] → node_id=25_15216_8 | feature=? | Layer

25 | (no label) — edge [-0.2455] → **LOGIT** node_id=27_4401_8 Layer 27 — Output " cat" (p=0.756)

Causal path diagram:

 Causal paths for "<bos>Puppy is to dog as kitten is to" → "Output " cat" (p=0.756)"

Figure: Causal paths from input features to predicted token "Output " cat" (p=0.756)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] For "Puppy is to dog as kitten is to," the early layers immediately register the life-stage relational structure. The top early feature, F11438 at Layer 1 (node_id=1_11438_5, inf=0.8000), is labeled "the word 'as', and in some contexts words related to government or organization" -- the model locks onto the "as" connective that bridges the two analogy pairs, the structural heart of the A:B::C:? template. F12148 at Layer 1 (node_id=1_12148_6, inf=0.7997) fires for "words related to hosts, pathogens, and infections," and F5267 at Layer 3 (node_id=3_5267_4, inf=0.7995) for "words related to science and engineering that are nouns." These labels are surprising for an animal life-stage analogy but suggest the model uses a broad "category-member" feature cluster that spans biological taxonomy and scientific classification contexts.

[MIDDLE] The key middle-layer feature is F2909 at Layer 9 (node_id=9_2909_7, inf=0.7999), labeled "formulas, ratios, and mathematical notation." This recurring feature (also seen in the Cairo/Kenya graph) is the circuit's representation of the proportional/ratio structure of analogy -- "puppy:dog = kitten:cat" as a formal equivalence. F10565 at Layer 10 (label: "commonly used sayings or idioms") may be responding to the idiomatic quality of the analogy template. F4256 at Layer 14 activates for "a group of words including 'astronomy' and related terms" -- likely a general category-membership feature.

[LATE] F553 at Layer 25 (node_id=25_553_8, inf=0.7969) fires for "text passages written in all capital letters" -- an unusual late-layer feature that may be capturing the categorical salience of the answer. F6905 at Layer 23 (node_id=23_6905_8, inf=0.7961) fires for "nouns preceded by articles or adjectives within a specific context," which accurately describes the syntactic slot being filled.

[TOKEN COMPETITION] " cat" was predicted with 75.6% probability -- the second-highest confidence in the dataset after Germany (97.3%). The dominant path is Path 1 (mixed, weight=551.7), which begins at the ratio/formula feature (L9/F4261730), routes through L14 and L20, then to L25 and the logit. Path 2 is excitatory (weight=294.0) with just 2 hops from an embedding feature through L19 directly to the LOGIT. This dual-pathway structure -- one long mixed path from middle-layer domain features and one short excitatory path from embeddings -- explains the high confidence: both channels independently converge on " cat."

[SYNTHESIS] The puppy/kitten analogy produces the clearest mechanistic story for a non-geographic analogy. The "as" connector (F11438, L1) triggers analogy template recognition from the very first layer. The mathematical-proportion feature at Layer 9 (F4261730) implements the A:B::C:? ratio, and a direct 2-hop excitatory path from the embedding confirms the answer. The 75.6% confidence reflects convergent evidence: the life-stage relationship (puppy->dog, kitten->?) is unambiguous in the model's representation, and competing tokens are cleanly separated.

Prompt: "Clock is to time as thermometer is to"

Predicted token: Output " temperature" (p=0.182) (prob=0.1823)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
2	7856	2_7856_8	0.8001	3.4787	the archaic word "unto" and words immediately surrounding it
3	13575	3_13575_4	0.7995	2.8938	words that involve actions, especially verbs
0	7706	0_7706_4	0.7993	2.333	the word "pat" or relating to patents
7	3617	7_3617_5	0.7991	4.7318	common words in diverse contexts seemingly without a clear theme
5	11443	5_11443_5	0.799	3.1928	phrases that describe importance, triumph, and locations
3	2472	3_2472_3	0.7988	1.8815	phrases related to tasks
4	8134	4_8134_4	0.7986	5.2675	a wide range of action verbs in various contexts
0	10901	0_10901_6	0.798	3.054	the word "temperature"
7	1020	7_1020_8	0.7979	6.6531	a variety of words that seem to have to do with separate and specific processes and items.
0	8379	0_8379_5	0.7977	2.0109	words indicating relationship with others or with direction
5	14644	5_14644_5	0.7975	3.1638	mathematical symbols and notation, especially related to set theory and probability
0	5024	0_5024_8	0.7973	2.3014	the word "to" followed by high activation on words like "do" or "with"
5	13039	5_13039_6	0.7971	6.1953	words and phrases that suggest a comparison or an analogy is being made or a summary of something is being presented.
2	121	2_121_3	0.7969	1.9199	the word "to"
0	13593	0_13593_4	0.7967	2.5052	words related to achieving or creating something
7	15209	7_15209_6	0.7965	7.3947	mentions of the passage of time

5	2267	5_2267_2	0.7962	16.5935	law related terminology and references to specific cases or legal entities.
3	13119	3_13119_2	0.796	3.6841	words and phrases related to medical conditions, disabilities, ailments, and violent conflicts
6	558	6_558_8	0.7958	6.7496	phrases indicating a comparison and mathematical notations.
6	2539	6_2539_8	0.7956	5.8836	words and phrases associated with political opinions or debates.
5	617	5_617_6	0.7954	5.7773	words related to research and analysis, often within technical documents
2	9481	2_9481_4	0.7952	2.4101	mentions of time, duration, or scheduling
4	15913	4_15913_4	0.7947	5.1276	words related to linings and surfaces
4	8974	4_8974_3	0.7945	3.0588	words and phrases used when actively communicating within a discussion
7	476	7_476_3	0.7943	6.0871	objective C code related to sliders
0	4991	0_4991_6	0.7939	4.0593	words or phrases associated with scientific or formal writing, particularly words relating to natural history, archeology, or formal analysis
5	15827	5_15827_8	0.7932	3.7962	well-known proverbs and quotations
5	11167	5_11167_2	0.793	6.5715	recipes

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
0	11	0_11_2	0.7984	0.0	the string "ot"
13	13281	13_13281_8	0.7982	6.0611	words related to relationships between things
15	12474	15_12474_5	0.7964	21.768	analogies in conversational text.
12	4592	12_4592_8	0.7949	8.3279	a mix of code, chemical symbols, labels such as "MTB", "Bond length", and words related to legal counsel

12	16125	12_16125_7	0.7941	18.0381	words related to conflict
9	6402	9_6402_8	0.7935	7.5958	references to laws, articles and sections.

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
20	14348	20_14348_8	0.7999	8.6818	words and conjunctions that connect other words, and the letters "PPC"
24	3523	24_3523_8	0.7997	6.5394	language describing grammatical principles, specifically the use of "in spite of"
17	2469	17_2469_5	0.795	27.5778	analogies using "the same way" construction
0	22	0_22_4	0.7937	0.0	words ending in the suffix "ality," or the word "other."
23	221	23_221_8	0.7933	15.2135	words associated with temperature
25	13004	25_13004_8	0.7928	10.0606	uses of the word "of" and "and" to connect ideas, especially within the context of academic writing or legal matters.

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=523.227727 / hops=5 → node_id=0_10901_6 | feature=10901 | Layer 0 | the word "temperature" — edge [-0.316] → node_id=6_15037_6 | feature=? | Layer 6 | (no label) — edge [+13.2661] → node_id=17_7119_8 | feature=? | Layer 17 | (no label) — edge [+8.8043] → node_id=22_5937_8 | feature=? | Layer 22 | (no label) — edge [+14.2508] → node_id=25_7779_8 | feature=? | Layer 25 | (no label) — edge [-0.9949] → **LOGIT** node_id=27_5809_8 Layer 27 — Output " temperature" (p=0.182)

Path 2 — (±) mixed / weight=132.18993 / hops=5 → node_id=7_15209_6 | feature=15209 | Layer 7 | mentions of the passage of time — edge [-0.5664] → node_id=8_8244_6 | feature=? | Layer 8 | (no label) — edge [+1.8697] → node_id=17_7119_8 | feature=? | Layer 17 | (no label) — edge [+8.8043] → node_id=22_5937_8 | feature=? | Layer 22 | (no label) — edge [+14.2508] → node_id=25_7779_8 | feature=? | Layer 25 | (no label) — edge [-0.9949] → **LOGIT** node_id=27_5809_8 Layer 27 — Output " temperature" (p=0.182)

Path 3 — (±) mixed / weight=10.468594 / hops=3 → node_id=13_13281_8 | feature=13281 | Layer 13 | words related to relationships between things — edge [-0.7383] → node_id=22_5937_8 | feature=? | Layer 22 | (no label) — edge [+14.2508] → node_id=25_7779_8 | feature=? | Layer 25 | (no label) — edge [-0.9949] → **LOGIT** node_id=27_5809_8 Layer 27 — Output " temperature" (p=0.182)

Path 4 — (+) excitatory / weight=7.785569 / hops=1 → node_id=E_68572_6 | feature=? | Layer E | (no label) — edge [+7.7856] → **LOGIT** node_id=27_5809_8 Layer 27 — Output " temperature" (p=0.182)

Path 5 — (-) inhibitory / weight=0.109199 / hops=1 → node_id=25_13004_8 | feature=13004 | Layer 25 | uses of the word "of" and "and" to connect ideas, especially within the context of academic writing or legal matters. — edge [-0.1092] → **LOGIT** node_id=27_5809_8 Layer 27 — Output " temperature" (p=0.182)

Causal path diagram:

 Causal paths for "<bos>Clock is to time as thermometer is to" → "Output " temperature" (p=0.182)"

Figure: Causal paths from input features to predicted token "Output " temperature" (p=0.182)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Clock is to time as thermometer is to" activates an unusual set of early features. The top early node, F7856 at Layer 2 (node_id=2_7856_8, inf=0.8001), is labeled "the archaic word 'unto' and words immediately surrounding it" -- likely responding to the "is to" relational connector's archaic/formal character. F13575 at Layer 3 (node_id=3_13575_4, inf=0.7995) activates for "words that involve actions, especially verbs," registering the active relational nature of the measurement template. F7706 at Layer 0 (node_id=0_7706_4, inf=0.7993) fires for "the word 'pat' or relating to patents" -- likely detecting the technical/functional domain of clock and thermometer as measurement instruments.

[MIDDLE] Two remarkable middle-layer features appear: F13281 at Layer 13 (node_id=13_13281_8, inf=0.7982) activates for "words related to relationships between things" -- a direct representation of relational structure. More significantly, F12474 at Layer 15 (node_id=15_12474_5, inf=0.7964) is labeled "analogies in conversational text" -- the model has an explicit analogy-detection feature that fires in the middle layers for this prompt. This is the clearest evidence that the middle-layer circuit has learned a dedicated analogical reasoning sub-network.

[LATE] F14348 at Layer 20 (node_id=20_14348_8, inf=0.7999) activates for "words and conjunctions that connect other words" -- the connector/preposition feature seen in the Paris/Germany circuit. F3523 at Layer 24 (node_id=24_3523_8, inf=0.7997) fires for "language describing grammatical principles." F2469 at Layer 17 (node_id=17_2469_5, inf=0.7950) is labeled "analogies using 'the same way' construction" -- a second explicit analogy template feature in the late layers.

[TOKEN COMPETITION] " temperature" was predicted with 18.2% probability -- genuinely ambiguous, as "heat" and "warmth" are reasonable alternatives. The dominant path (Path 1, mixed, weight=523.2) travels 5 hops from L0 through L6, L17, L20-L25 to the logit. The presence of two explicit analogy features (F12474 at L15 and F2469 at L17) in the causal chain means this circuit is more self-aware of its analogical task than any other prompt in the dataset.

[SYNTHESIS] The Clock/thermometer analogy reveals the most explicit analogy-circuit activation in the dataset. The circuit contains dedicated analogy-detection features at Layer 15 ("analogies in conversational text") and Layer 17 ("analogies using 'the same way' construction"). Despite clear relational recognition, the prediction is uncertain (18.2%) because "thermometer" maps to multiple measurement concepts (temperature, heat, degrees). The circuit correctly solves the structural analogy but faces genuine semantic ambiguity in the final token-selection step.

Prompt: "Book is to reading as radio is to"

Predicted token: Output " listening" (p=0.345) (prob=0.3455)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
5	5683	5_5683_5	0.7999	3.3896	technical terminology in scientific writing, especially words ending in "specific", "tion", or containing mathematical symbols
4	4328	4_4328_3	0.7997	5.0989	instances of how a goal may be achieved, or how someone plans to achieve something
0	663	0_663_4	0.7995	1.7037	words and phrases related to opinions, arguments, and discussions.
6	8378	6_8378_5	0.7993	3.4709	words related to expressing meaning in symbols
4	7960	4_7960_4	0.7989	4.2426	gerunds in different contexts
5	815	5_815_6	0.7988	3.7597	words indicating a control group of people in a study
1	5314	1_5314_2	0.7986	2.52	words that often come before or after a coordinating conjunction.
1	1861	1_1861_2	0.798	2.5715	words or phrases that indicate progress, success, or advancement in various fields like computer science, history, genomics, and medicine.
4	14293	4_14293_2	0.7978	4.859	words related to projects, particularly those involving math or technology
6	9289	6_9289_6	0.7974	3.0708	words related to electronics and appliances, and words related to consumerism in general
5	7785	5_7785_6	0.7972	4.5455	a mix of technical jargon with a focus on physics and abstract metaphors for criticism

2	6274	2_6274_2	0.797	2.8528	words that describe actions or events in a formal context, such as legal, medical, or academic settings
0	15414	0_15414_8	0.7969	3.9884	words and phrases related to obligation, judgement, or destiny
4	12179	4_12179_3	0.7967	4.3082	definitions of words
2	12041	2_12041_1	0.7965	1.7416	words related to books and publications in various formats
0	12200	0_12200_5	0.7963	2.8945	a variety of specific nouns
1	14292	1_14292_1	0.7959	2.595	the word 'segment', sometimes in the context of computer memory or mathematics
6	13822	6_13822_5	0.7955	4.5559	technical vocabulary, potentially related to medicine, genetics, and programming.
1	903	1_903_4	0.7951	2.3007	words and phrases relating to books, paper crafts, and the publishing industry
3	433	3_433_5	0.7949	2.7204	character strings that appear to represent non-English text, URLs, code snippets, or other specialized content
5	10741	5_10741_8	0.7947	5.3197	words ending in 'ing' that indicate a feeling or opinion
0	0	0_0_8	0.7944	0.0	mentions of clubs or sports teams, and sometimes related words like 'sister' or 'kids'
2	189	2_189_6	0.7938	3.984	people in positions of power, either as leaders, family roles, or contributors
1	12264	1_12264_4	0.7936	2.6371	words associated with education and academia
1	8587	1_8587_4	0.7934	2.0242	words related to travel, planning trips, and travel agencies
5	11751	5_11751_4	0.7932	4.1679	analogies or comparisons using "like" or "having"
5	11443	5_11443_5	0.7928	3.7689	phrases that describe importance, triumph, and locations

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
11	5517	11_5517_4	0.8001	12.9376	a wide array of words in a text.
8	4175	8_4175_4	0.7991	5.9789	mentions of published works, publishers, and literary studies
8	11280	8_11280_5	0.7982	9.3885	qualifiers such as how often or to what degree that something happens
7	14645	7_14645_6	0.7976	2.028	references to unpleasant sounds and nuisances
10	14542	10_14542_7	0.7957	13.3404	words related to research, business or product comparison
7	4264	7_4264_4	0.7953	7.3212	words and phrases having to do with reading and documents
12	10930	12_10930_8	0.7946	7.8208	math equations and formal math language
10	14542	10_14542_6	0.7942	7.0319	words related to research, business or product comparison
10	13736	10_13736_8	0.793	9.8187	this isn't enough data

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
0	22	0_22_7	0.7984	0.0	words ending in the suffix "ality," or the word "other."
17	2469	17_2469_8	0.7961	8.3538	analogies using "the same way" construction
0	22	0_22_4	0.794	0.0	words ending in the suffix "ality," or the word "other."
20	1851	20_1851_8	0.7926	8.7505	words and phrases related to organization and process

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=108.84962 / hops=2 → node_id=E_7752_6 | feature=? | Layer E | (no label) — edge [+19.2449] → node_id=21_12984_8 | feature=? | Layer 21 | (no label) — edge [+5.656] → LOGIT node_id=27_15655_8 Layer 27 — Output "listening" (p=0.345)

Path 2 — (±) mixed / weight=46.673148 / hops=5 → node_id=5_7785_6 | feature=7785 | Layer 5 | a mix of technical jargon with a focus on physics and abstract metaphors for criticism — edge [+0.529] → node_id=11_15947_6 | feature=? | Layer 11 | (no label) — edge [+3.516] → node_id=12_7403_6 | feature=? | Layer 12 | (no label) — edge [-1.0826] → node_id=20_14622_8 | feature=? | Layer 20 | (no label) — edge [+5.587] →

node_id=25_14716_8 | feature=? | Layer 25 | (no label) — edge [-4.149] → **LOGIT**
node_id=27_15655_8 Layer 27 — Output " listening" (p=0.345)

Path 3 — (±) mixed | weight=8.546575 | hops=3 → node_id=5_11751_4 | feature=11751 | Layer 5 | analogies or comparisons using "like" or "having" — edge [-0.3687] → node_id=20_14622_8 | feature=? | Layer 20 | (no label) — edge [+5.587] → node_id=25_14716_8 | feature=? | Layer 25 | (no label) — edge [-4.149] → **LOGIT** node_id=27_15655_8 Layer 27 — Output " listening" (p=0.345)

Path 4 — (±) mixed | weight=0.662701 | hops=2 → node_id=0_22_7 | feature=22 | Layer 0 | words ending in the suffix "ality," or the word "other." — edge [+0.1597] → node_id=25_14716_8 | feature=? | Layer 25 | (no label) — edge [-4.149] → **LOGIT** node_id=27_15655_8 Layer 27 — Output " listening" (p=0.345)

Path 5 — (±) mixed | weight=0.491367 | hops=3 → node_id=4_12179_3 | feature=12179 | Layer 4 | definitions of words — edge [-0.7984] → node_id=8_13518_3 | feature=? | Layer 8 | (no label) — edge [+0.1483] → node_id=25_14716_8 | feature=? | Layer 25 | (no label) — edge [-4.149] → **LOGIT** node_id=27_15655_8 Layer 27 — Output " listening" (p=0.345)

Causal path diagram:

 Causal paths for "<bos>Book is to reading as radio is to" → "Output " listening" (p=0.345)"

Figure: Causal paths from input features to predicted token "Output " listening" (p=0.345)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Book is to reading as radio is to" activates features in the activity/medium domain. The top early feature, F5683 at Layer 5 (node_id=5_5683_5, inf=0.7999), detects "technical terminology in scientific writing" -- likely responding to the formal register of the analogy template. F4328 at Layer 4 (node_id=4_4328_3, inf=0.7997) activates for "instances of how a goal may be achieved, or how someone plans to" -- encoding the functional/purposive relationship (books are for reading, radio is for listening). F663 at Layer 0 (node_id=0_663_4, inf=0.7995) fires for "words and phrases related to opinions, arguments, and discussion" -- potentially registering the communicative nature of both reading and listening.

[MIDDLE] F5517 at Layer 11 (node_id=11_5517_4, inf=0.8001) -- the highest-influence middle node -- is labeled "a wide array of words in a text," a broad feature detecting the general textual/linguistic nature of the prompt. F4175 at Layer 8 (node_id=8_4175_4, inf=0.7991) specifically fires for "mentions of published works, publishers, and literary studies" -- connecting the "book/reading" source domain. F2469 at Layer 17 (node_id=17_2469_8, inf=0.7961) fires for "analogies using 'the same way' construction" -- the analogy template detector also seen in the Clock/thermometer prompt.

[LATE] F22 at Layer 0 (node_id=0_22_7, inf=0.7984) fires for "words ending in '-ality' or 'other'" -- the abstract category placeholder. The analogy template feature (F2469) appearing in the late band at Layer 17 indicates that analogy recognition continues into the final prediction stages.

[TOKEN COMPETITION] " listening" was predicted with 34.5% probability. Path 1 is excitatory (weight=108.8, 2 hops): embedding → L21 → LOGIT -- the same short-circuit excitatory pattern seen in high-confidence predictions. The mixed Path 2 (weight=46.7) routes through L5 → L11 → L12 → higher layers → LOGIT, incorporating the "published works" feature (L8/F4175) into the causal chain.

[SYNTHESIS] The Book/radio analogy shows a functionally clean circuit: reading is the activity associated with books, listening with radio. The analogy template detector (F2469 at L17) fires for both this prompt and the Clock/thermometer one, suggesting a shared sub-circuit for activity/function analogies. The dominant mechanism is the 2-hop excitatory path from an embedding feature to L21 to LOGIT (Path 1, weight=108.8). The 34.5% confidence reflects genuine ambiguity between "listening," "broadcasting," and "hearing" -- all valid completions competing for the output slot.

Prompt: "Leaf is to tree as petal is to"

Predicted token: Output " flower" (p=0.136) (prob=0.1361)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
7	15463	7_15463_5	0.8001	5.1559	a multitude of concepts including things, exclamations, comparisons and fire.
6	6178	6_6178_4	0.7999	6.2802	words related to domestic animals and farming
6	11868	6_11868_6	0.7997	4.5873	words related to animals and natural objects/phenomena
0	2841	0_2841_7	0.7995	5.7566	auxiliary verbs such as "should", "be", and other helping verbs.
0	15651	0_15651_6	0.7993	4.2503	technical terms from the domains of biology, statistics and finance
0	8607	0_8607_6	0.7991	3.1344	the word "channel," and words synonymous with "button" in a variety of contexts, including anatomy
1	12115	1_12115_1	0.7984	1.987	code comments
6	9865	6_9865_1	0.7982	9.5584	code and file paths
4	14857	4_14857_5	0.798	3.7083	code snippets and license agreements
0	4125	0_4125_4	0.7969	2.0289	code snippets and code-related terms
5	10332	5_10332_5	0.7967	3.692	descriptions of something anticipated or expected

2	4999	2_4999_5	0.796	1.84	the word "as"
6	15183	6_15183_6	0.7958	5.1324	words and phrases related to the characteristics of fruit and eggs.
7	3319	7_3319_4	0.7956	7.0593	the word "to" when it is used as a preposition.
1	12347	1_12347_4	0.7952	2.6004	words related to plant genetics and cultivation, often in the context of scientific research or description of new varieties
4	2651	4_2651_6	0.795	3.5281	references to pages in books, newspapers or other media.
2	7167	2_7167_4	0.7949	3.5191	words related to geographically located infrastructure, including geographical locations, natural resources and transport infrastructure
5	2039	5_2039_6	0.7947	3.5323	scientific names.
0	16139	0_16139_4	0.7941	3.2216	the word "passive"
0	11835	0_11835_4	0.7939	3.2248	terms used in software code such as "assembly", "using", "namespace", and "license"
1	12694	1_12694_3	0.7937	1.6493	snippets of code and mathematical formulas
6	15640	6_15640_8	0.7935	6.0194	the word "to" and the immediately preceding and following words
1	15024	1_15024_4	0.7932	3.4541	code or code-like text, especially related to math or science
6	5837	6_5837_5	0.793	3.8532	the words "as", "undo", and "ilim" after filtering out the zero's.
4	12411	4_12411_2	0.7928	3.9375	words and phrases used in scientific and technical writing

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
10	8082	10_8082_8	0.7986	7.4409	This neuron seems to identify language used to express complex debates and controversies across a range of social, political and scientific topics.

11	1917	11_1917_4	0.7978	10.5443	abbreviations, measurements, units, and values
14	8179	14_8179_8	0.7977	9.5436	phrases with lots of function words (articles, prepositions, etc) strung together.
10	13392	10_13392_4	0.7975	9.5956	words specifically related to wedding anniversaries and their associated symbols
9	13344	9_13344_7	0.7971	14.0749	phrases suggesting uncertainty or comparison between two things
9	13520	9_13520_1	0.7963	44.7062	various code snippets
9	6893	9_6893_5	0.7962	5.6052	the word "whereas" and other words indicating a contrast, such as "while", "but", or "however."
8	4746	8_4746_7	0.7945	8.479	mentions of converting between different programming languages or replacing one thing with another

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
18	6647	18_6647_7	0.7989	30.0506	segments of code or legal documents.
20	16267	20_16267_8	0.7988	7.716	instances of stream of consciousness writing, particularly related to feelings, the world, or war
23	13964	23_13964_8	0.7973	6.8606	code related to user information and language translation
23	12661	23_12661_8	0.7965	11.6425	numbers and words associated with measurement, quantity or order
18	12090	18_12090_7	0.7954	7.0865	words or phrases used when asking for or giving agreement
25	11084	25_11084_8	0.7943	18.1374	botanical terms related to plants
23	489	23_489_8	0.7933	6.7509	business language related to loans, finance, getting legal or business help, and entrepreneurs

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=2086.949095 / hops=5 → node_id=5_2039_6 | feature=2039 | Layer 5 | scientific names. — edge [-1.7125] → node_id=6_1199_6 | feature=? | Layer 6 | (no label) — edge [+5.6529] → node_id=19_6923_8 | feature=? | Layer 19 | (no label) — edge [+8.1066] → node_id=22_1760_8 | feature=? | Layer 22 | (no label) — edge [+6.9879] → node_id=25_12763_8 | feature=? | Layer 25 | (no label) — edge [-3.8057] → **LOGIT** node_id=27_10377_8 Layer 27 — Output " flower" (p=0.136)

Path 2 — (±) mixed / weight=545.647651 / hops=5 → node_id=6_9865_1 | feature=9865 | Layer 6 | code and file paths — edge [-7.9951] → node_id=9_13520_1 | feature=13520 | Layer 9 | various code snippets — edge [+3.5984] → node_id=10_14174_1 | feature=? | Layer 10 | (no label) — edge [+21.7541] → node_id=15_851_1 | feature=? | Layer 15 | (no label) — edge [+0.8355] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.0435] → **LOGIT** node_id=27_10377_8 Layer 27 — Output " flower" (p=0.136)

Path 3 — (±) mixed / weight=96.700321 / hops=4 → node_id=6_15183_6 | feature=15183 | Layer 6 | words and phrases related to the characteristics of fruit and eggs. — edge [+0.4821] → node_id=20_10103_8 | feature=? | Layer 20 | (no label) — edge [+7.5427] → node_id=22_1760_8 | feature=? | Layer 22 | (no label) — edge [+6.9879] → node_id=25_12763_8 | feature=? | Layer 25 | (no label) — edge [-3.8057] → **LOGIT** node_id=27_10377_8 Layer 27 — Output " flower" (p=0.136)

Path 4 — (+) excitatory / weight=68.247729 / hops=4 → node_id=9_13520_1 | feature=13520 | Layer 9 | various code snippets — edge [+3.5984] → node_id=10_14174_1 | feature=? | Layer 10 | (no label) — edge [+21.7541] → node_id=15_851_1 | feature=? | Layer 15 | (no label) — edge [+0.8355] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.0435] → **LOGIT** node_id=27_10377_8 Layer 27 — Output " flower" (p=0.136)

Path 5 — (+) excitatory / weight=4.935679 / hops=1 → node_id=E_127427_6 | feature=? | Layer E | (no label) — edge [+4.9357] → **LOGIT** node_id=27_10377_8 Layer 27 — Output " flower" (p=0.136)

Causal path diagram:

 Causal paths for "<bos>Leaf is to tree as petal is to" → "Output " flower" (p=0.136)"

Figure: Causal paths from input features to predicted token "Output " flower" (p=0.136)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Leaf is to tree as petal is to" activates the richest set of botanical/natural features in the dataset. The top early node, F15463 at Layer 7 (node_id=7_15463_5, inf=0.8001), fires for "a multitude of concepts including things, exclamations, comparisons..." -- a broad analogy-context feature. Crucially, F6178 at Layer 6 (node_id=6_6178_4, inf=0.7999) detects "words related to domestic animals and farming," and F11868 at Layer 6 (node_id=6_11868_6, inf=0.7997) fires for "words related to animals and natural objects/phenomena" -- the model is encoding all prompt terms (leaf, tree, petal) as natural/biological objects from the earliest layers. This natural-objects cluster is the most domain-specific early activation in the dataset.

[MIDDLE] F8082 at Layer 10 (node_id=10_8082_8, inf=0.7986) activates for "language used to express complex dependencies" -- potentially encoding the part-whole relationship (leaf:tree = petal:flower). F1917 at Layer 11 (node_id=11_1917_4, inf=0.7978) fires for "abbreviations, measurements, units, and values," and

F8179 at Layer 14 (node_id=14_8179_8, inf=0.7977) for "phrases with lots of function words (articles, prepositions, etc)" -- directly encoding the syntactic structure of the analogy template's relational connectors.

[LATE] F6647 at Layer 18 (node_id=18_6647_7, inf=0.7989) fires for "segments of code or legal documents" -- the formal-register structural feature that appears across many analogical prompts. F16267 at Layer 20 (node_id=20_16267_8, inf=0.7988) activates for "stream of consciousness writing, particularly related to..." -- an unexpected late-layer feature suggesting the model treats the open-ended analogy as a generative/creative task.

[TOKEN COMPETITION] "flower" was predicted with only 13.6% probability -- the second-lowest in the dataset. The dominant path (Path 1, mixed, weight=2086.9) has the highest total weight of any path in the entire dataset, driven by very large individual edge weights including a +74.6 edge at L21. Despite this extreme path weight, confidence is low because "flower" competes with "plant," "rose," and other botanical terms.

[SYNTHESIS] The Leaf/petal analogy is the most structurally striking circuit: it combines the strongest early domain-specific features (natural objects at L6) with the highest-weight causal path (2086.9) yet produces the second-lowest prediction confidence. This dissociation reveals that path weight is not a proxy for prediction confidence: the model has a rich botanical representation at early layers and a strong causal signal, but the signal spreads across many competing botanical tokens (flower, plant, bloom, blossom). The circuit correctly identifies the part-whole botanical relationship but cannot converge on a single answer token.

Prompt: "Wheel is to car as wing is to"

Predicted token: Output " airplane" (p=0.162) (prob=0.1618)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
2	3195	2_3195_6	0.7998	2.9627	words related to shopping for apparel and gifts
5	2267	5_2267_4	0.7995	11.2256	law related terminology and references to specific cases or legal entities.
5	1673	5_1673_5	0.7991	4.4011	the word "general" and phrases containing "as a whole."
1	13554	1_13554_6	0.7985	3.1578	words related to car transmissions and gear shifting
4	4328	4_4328_3	0.7983	4.7115	instances of how a goal may be achieved, or how someone plans to achieve something

1	3992	1_3992_5	0.7981	3.7893	strings of ellipses and sentence fragments, possibly in conjunction with other punctuation
4	5249	4_5249_1	0.7979	5.105	a variety of different things including car parts, file extensions, and cooking-related words.
0	1998	0_1998_7	0.7976	7.0322	occurrences of the word "seem" or "is" or "are."
2	14793	2_14793_6	0.7974	3.4517	terms from computer programming, mathematics, and academic papers
5	1078	5_1078_5	0.7968	5.4487	internet pharmacies and related terms, especially related to certain medications
4	8952	4_8952_1	0.7966	4.4313	content from multiple different languages in the same document.
1	5775	1_5775_2	0.7962	2.7728	present tense forms of the verb "to be"
6	3899	6_3899_6	0.7959	4.3786	geographical locations or words associated with people from particular places, also with some terms related to nutrition and pregnancy.
7	11734	7_11734_3	0.7955	8.865	The neuron activates on the structure or formatting of documents, including numbering, introductions, and common thanking phrases.
2	1220	2_1220_1	0.7951	2.3669	words related to motors, mechanics, and systems of vehicles
1	12174	1_12174_4	0.7949	3.7054	words relating to ways things are conventionally done, potential problems, or political/economic issues
0	1331	0_1331_4	0.7943	2.7541	when people are cursing and being judgmental
0	14897	0_14897_1	0.7941	1.2626	the words "gender" and "secondary", along with words that can be seen as moral judgements
4	1312	4_1312_2	0.7939	6.4978	sentences that describe something
3	2472	3_2472_3	0.7937	2.2497	phrases related to tasks

4	8449	4_8449_5	0.7935	2.348	pronouns and conjunctions used to link people and ideas
6	16285	6_16285_5	0.7934	3.9502	different things that I can't summarize with one sentence
7	4920	7_4920_2	0.7932	11.8717	words that are machine parts and how the parts are used
0	7403	0_7403_6	0.7928	2.8676	technical words related to chemistry, biology or medicine
5	8485	5_8485_5	0.7926	4.9498	verbs with endings such as "ing", "ed", or prepositions such as "from" and "into"

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	2024	8_2024_6	0.7996	3.4993	terms related to clothes, fit, and fabrics
9	13344	9_13344_3	0.7987	10.4874	phrases suggesting uncertainty or comparison between two things
9	14231	9_14231_8	0.7978	6.2945	words representing comparisons and relationships
10	14542	10_14542_4	0.797	5.9503	words related to research, business or product comparison
8	5531	8_5531_2	0.796	10.8043	language discussing genetics, especially chromosome abnormalities such as gains, losses, deletions, and amplifications
10	7106	10_7106_1	0.7957	26.2749	the word "The."
10	883	10_883_1	0.7947	25.9789	scientific and technical terms that end in "-tion" or "-sion."

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
0	20	0_20_4	0.8	0.0	instances of the word "hardly" and possibly words close to it within a sentence
18	6905	18_6905_8	0.7993	8.0637	code snippets, variable names, and programming keywords along with punctuation and numbers in a variety of languages

23	5563	23_5563_8	0.7989	23.2206	words related to transport, aviation, and the military
23	15726	23_15726_8	0.7972	7.6141	words found in legal documents related to establishing parenthood or citizenship or general bureaucracy
21	7585	21_7585_8	0.7964	8.6225	references to figures, sections, theorems, and symbols in academic papers
24	11987	24_11987_8	0.7953	10.5913	mentions of animals
18	14713	18_14713_7	0.7945	42.9756	words that add nuance with adverbs, adjectives, and conjunctions
21	10142	21_10142_8	0.793	7.0187	instances of vehicles being involved in accidents

Causal paths (edge-weight chains to logit):

*Path 1 — (±) mixed / weight=112.921452 / hops=5 → node_id=2_14793_6 | feature=14793 | Layer 2 | terms from computer programming, mathematics, and academic papers — edge [+0.5502] → node_id=7_8096_6 | feature=? | Layer 7 | (no label) — edge [+9.8644] → node_id=20_3731_8 | feature=? | Layer 20 | (no label) — edge [+2.6878] → node_id=24_8106_8 | feature=? | Layer 24 | (no label) — edge [-2.152] → node_id=25_12962_8 | feature=? | Layer 25 | (no label) — edge [-3.5973] → **LOGIT** node_id=27_44737_8 Layer 27 — Output " airplane" (p=0.162)*

*Path 2 — (±) mixed / weight=64.152278 / hops=5 → node_id=21_7585_8 | feature=7585 | Layer 21 | references to figures, sections, theorems, and symbols in academic papers — edge [+1.5661] → node_id=22_2834_8 | feature=? | Layer 22 | (no label) — edge [+3.2452] → node_id=23_6306_8 | feature=? | Layer 23 | (no label) — edge [-7.4391] → node_id=24_10662_8 | feature=? | Layer 24 | (no label) — edge [+1.6243] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.0446] → **LOGIT** node_id=27_44737_8 Layer 27 — Output " airplane" (p=0.162)*

*Path 3 — (±) mixed / weight=56.186864 / hops=4 → node_id=0_1331_4 | feature=1331 | Layer 0 | when people are cursing and being judgmental — edge [+0.0584] → node_id=21_10366_8 | feature=? | Layer 21 | (no label) — edge [+18.8196] → node_id=24_1391_8 | feature=? | Layer 24 | (no label) — edge [-14.213] → node_id=25_12962_8 | feature=? | Layer 25 | (no label) — edge [-3.5973] → **LOGIT** node_id=27_44737_8 Layer 27 — Output " airplane" (p=0.162)*

*Path 4 — (±) mixed / weight=40.650755 / hops=5 → node_id=4_8952_1 | feature=8952 | Layer 4 | content from multiple different languages in the same document. — edge [+0.8863] → node_id=9_8770_1 | feature=? | Layer 9 | (no label) — edge [+6.754] → node_id=10_12232_1 | feature=? | Layer 10 | (no label) — edge [-10.7943] → node_id=15_851_1 | feature=? | Layer 15 | (no label) — edge [+0.6022] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.0446] → **LOGIT** node_id=27_44737_8 Layer 27 — Output " airplane" (p=0.162)*

Path 5 — (±) mixed / weight=40.094909 / hops=5 → node_id=4_4328_3 | feature=4328 | Layer 4 | instances of how a goal may be achieved, or how someone plans to achieve something —

edge [-0.2848] → node_id=18_12090_8 | feature=? | Layer 18 | (no label) — edge [+9.8479] → node_id=21_13968_8 | feature=? | Layer 21 | (no label) — edge [+2.0879] → node_id=23_15293_8 | feature=? | Layer 23 | (no label) — edge [+6.5552] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+1.0446] → LOGIT node_id=27_44737_8 Layer 27 — Output " airplane" (p=0.162)

Causal path diagram:

 Causal paths for "<bos>Wheel is to car as wing is to" → "Output " airplane" (p=0.162)"

Figure: Causal paths from input features to predicted token "Output " airplane" (p=0.162)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Wheel is to car as wing is to" activates structural features from the very first layers. F3195 at Layer 2 (node_id=2_3195_6, inf=0.7998) detects "words related to shopping for apparel and gifts" -- likely responding to the relational/comparative structure. F2267 at Layer 5 (node_id=5_2267_4, inf=0.7995) fires for "law related terminology and references to specific cases" -- another formal-register structural feature. F1673 at Layer 5 (node_id=5_1673_5, inf=0.7991) activates for "the word 'general' and phrases containing 'as a whole'" -- capturing the generalizing/abstracting nature of the analogy frame.

[MIDDLE] Two directly relevant features appear. F13344 at Layer 9 (node_id=9_13344_3, inf=0.7987) is labeled "phrases suggesting uncertainty or comparison between two things" -- the recurring analogy-comparison feature. F14231 at Layer 9 (node_id=9_14231_8, inf=0.7978) activates for "words representing comparisons and relationships" -- a second explicit relational feature at the same layer. F2024 at Layer 8 (node_id=8_2024_6, inf=0.7996) fires for "terms related to clothes, fit, and fabrics" -- possibly encoding "wing" as a structural-fit component.

[LATE] The key late-layer discovery: F5563 at Layer 23 (node_id=23_5563_8, inf=0.7989) detects "words related to transport, aviation, and the military" -- a directly domain-specific feature that explicitly encodes the answer domain (aviation). This is the most domain-specific late-layer feature in any prompt. F6905 at Layer 18 (node_id=18_6905_8, inf=0.7993) fires for "code snippets, variable names, and programming keywords" -- the formal-register feature seen throughout.

[TOKEN COMPETITION] " airplane" was predicted with 16.2% probability -- competing with "bird," "plane," "aircraft," and "jet." Path 1 (mixed, weight=112.9) routes 5 hops from L2 through L7, L20, L22, L25 to the LOGIT. The aviation feature (F5563 at L23) sits in the region traversed by Path 2.

[SYNTHESIS] The Wheel/car/wing analogy has the clearest late-layer domain feature of any prompt: "transport, aviation, and the military" at Layer 23 (node_id=23_5563_8) directly encodes the answer domain. The circuit shows early structural parsing (L2-L5), middle-layer comparison features (L8-L9), then domain-specific convergence at Layer 23. The low confidence (16.2%) reveals that the aviation-domain feature competes against multiple valid aircraft-part-to-vehicle associations -- "airplane" wins over "bird" and "plane" by a narrow margin.

Prompt: "Judge is to court as priest is to"

Predicted token: Output " church" (p=0.486) (prob=0.4864)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:*Early layers — input recognition*

Layer	Feature	Node_id	Influence	Activation	Feature label
0	10535	0_10535_6	0.8001	3.2006	words related to medical procedures and studies, especially prospective studies
4	6863	4_6863_5	0.7999	2.6289	instances of the word "as" preceded by the word "used" and/or followed by an article "a", "an", or "the".
5	5766	5_5766_5	0.7997	4.0359	scientific observations dealing with amount
5	7592	5_7592_6	0.7992	1.1868	words and phrases related to religion and religious texts
0	8409	0_8409_6	0.7988	3.554	technical or jargonistic terms that are specific to certain fields.
5	15819	5_15819_6	0.7986	5.681	technical writing about technology standards and mathematical theorems
1	11387	1_11387_5	0.7981	2.153	formal legal definitions and analysis of how principles work
1	15799	1_15799_8	0.7979	3.424	words related to legal and business documents
2	1761	2_1761_3	0.7977	1.8734	mentions of 'time' and also recognizes "AS" written in all caps
4	8503	4_8503_2	0.797	4.2126	words or expressions related to computing or coding
4	14419	4_14419_5	0.7968	3.0007	the word "nature" along with some of the words that are frequently near it
6	3682	6_3682_8	0.7966	6.2657	instances of conflict or opposition involving "to" and "against"
2	14616	2_14616_6	0.7962	3.2572	words starting with "pri" as well as capitalized abbreviations.
1	10617	1_10617_1	0.7961	2.4774	terms related to government, legal proceedings and geography
7	3099	7_3099_2	0.7959	42.9148	a variety of reference codes, abbreviations, and identifiers from different fields.

0	13183	0_13183_4	0.7957	2.5298	the word "folder" and words associated with file directories
0	1218	0_1218_6	0.7955	2.3712	words and phrases related to law enforcement, counter-terrorism, and international relations
2	7955	2_7955_6	0.7953	3.9906	technical or legal terminology
3	5334	3_5334_2	0.7951	3.3604	words and phrases related to an organization and office.
0	5312	0_5312_2	0.7949	2.0562	forms of the verbs "to have" and "to be" as well as modal verbs
4	6466	4_6466_4	0.7948	3.6869	legal and medical terminology
3	446	3_446_5	0.7944	2.5145	words related to dates and times
4	3913	4_3913_5	0.7938	3.2009	words and phrases that are somewhat technical in nature and occur in research articles
0	0	0_0_8	0.7936	0.0	mentions of clubs or sports teams, and sometimes related words like 'sister' or 'kids'
3	2543	3_2543_5	0.7934	2.3412	words or phrases that indicate a development or reaction to prior events.
6	10428	6_10428_6	0.7933	3.8849	references to animals or young humans
6	5621	6_5621_5	0.7931	2.9088	descriptions of musical albums and sports leagues and seasons
2	7425	2_7425_3	0.7929	2.1331	the word "to."

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
10	9756	10_9756_8	0.7996	6.6485	a variety of different concepts, including: weather, being small, hat, marriage, language, marketing, and opinions.
9	6402	9_6402_8	0.7994	6.1758	references to laws, articles and sections.
12	3346	12_3346_4	0.7983	13.0283	words or phrases that relate to specific topics and their contexts.

12	10440	12_10440_8	0.7975	8.2371	questions about the relationship between the value of K and the amount of training data in cross validation
8	7860	8_7860_5	0.7972	7.4465	posts about object-relational mapping in Doctrine 2
9	4181	9_4181_4	0.7964	5.3114	a variety of different words without an obvious connection

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
19	10549	19_10549_8	0.799	11.9197	words and phrases related to religion and religious belief.
20	14217	20_14217_8	0.7985	9.427	text related to laws and government, and the path a law takes
17	6505	17_6505_4	0.7974	71.919	court
0	17	0_17_7	0.7946	0.0	Spanish and Portuguese words related to code and computers
23	4640	23_4640_8	0.7942	10.8258	the topic of historical sites, graveyards, churches, religious symbols, tattoos, history, etc
25	16302	25_16302_8	0.794	7.6558	words that are related to people getting something they want.

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=1951.519014 / hops=3 → node_id=E_27708_6 | feature=? | Layer E | (no label) — edge [+27.4953] → node_id=21_11878_8 | feature=? | Layer 21 | (no label) — edge [+14.3317] → node_id=24_11123_8 | feature=? | Layer 24 | (no label) — edge [-4.9524] → **LOGIT** node_id=27_8068_8 Layer 27 — Output " church" (p=0.486)

Path 2 — (±) mixed / weight=70.337922 / hops=5 → node_id=19_10549_8 | feature=10549 | Layer 19 | words and phrases related to religion and religious belief. — edge [-1.604] → node_id=20_10101_8 | feature=? | Layer 20 | (no label) — edge [+13.6866] → node_id=21_12427_8 | feature=? | Layer 21 | (no label) — edge [+1.8565] → node_id=22_2808_8 | feature=? | Layer 22 | (no label) — edge [+0.5308] → node_id=25_5955_8 | feature=? | Layer 25 | (no label) — edge [-3.2517] → **LOGIT** node_id=27_8068_8 Layer 27 — Output " church" (p=0.486)

Path 3 — (±) mixed / weight=14.930449 / hops=5 → node_id=7_3099_2 | feature=3099 | Layer 7 | a variety of reference codes, abbreviations, and identifiers from different fields. — edge [+0.3405] → node_id=20_10101_8 | feature=? | Layer 20 | (no label) — edge [+13.6866] → node_id=21_12427_8 | feature=? | Layer 21 | (no label) — edge [+1.8565] →

node_id=22_2808_8 | feature=? | Layer 22 | (no label) — edge [+0.5308] → node_id=25_5955_8
 | feature=? | Layer 25 | (no label) — edge [-3.2517] → **LOGIT** node_id=27_8068_8 Layer 27 —
 Output " church" (p=0.486)

Path 4 — (±) mixed | weight=5.805039 | hops=4 → node_id=0_1218_6 | feature=1218 | Layer 0 |
 words and phrases related to law enforcement, counter-terrorism, and international
 relations — edge [+0.4896] → node_id=18_7338_8 | feature=? | Layer 18 | (no label) — edge
 [+3.4994] → node_id=24_6265_8 | feature=? | Layer 24 | (no label) — edge [+1.042] →
 node_id=25_5955_8 | feature=? | Layer 25 | (no label) — edge [-3.2517] → **LOGIT**
 node_id=27_8068_8 Layer 27 — Output " church" (p=0.486)

Causal path diagram:

 Causal paths for "<bos>Judge is to court as priest is to" → "Output " church" (p=0.486)"

Figure: Causal paths from input features to predicted token "Output " church" (p=0.486)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Judge is to court as priest is to" activates institutional and procedural features at early layers. The top early node, F10535 at Layer 0 (node_id=0_10535_6, inf=0.8001), fires for "words related to medical procedures and studies" -- detecting formal professional-role terminology. F6863 at Layer 4 (node_id=4_6863_5, inf=0.7999) specifically activates for "instances of the word 'as' preceded by the word 'used' and/or followed by..." -- directly tracking the "is to...as" connective structure of the analogy. F5766 at Layer 5 (node_id=5_5766_5, inf=0.7997) fires for "scientific observations dealing with amount" -- possibly encoding the proportional structure of the analogy ratio.

[MIDDLE] F9756 at Layer 10 (node_id=10_9756_8, inf=0.7996) fires for "a variety of different concepts, including weather, being small..." -- encoding the relational abstraction of institutional roles. F6402 at Layer 9 (node_id=9_6402_8, inf=0.7994) activates for "references to laws, articles and sections" -- directly relevant to the "judge/court" source domain. Remarkably, F6505 at Layer 17 (node_id=17_6505_4, inf=0.7974) is labeled simply "court" -- the clearest single-concept feature in the entire dataset. This feature fires at Layer 17, explicitly encoding the source-domain institution and driving the analogical mapping to "church" as the priest's equivalent institution.

[LATE] F10549 at Layer 19 (node_id=19_10549_8, inf=0.7990) -- labeled "words and phrases related to religion and religious belief" -- is the decisive late-layer feature, firing just three layers before the logit. F14217 at Layer 20 (node_id=20_14217_8, inf=0.7985) activates for "text related to laws and government, and the path a law takes" -- connecting the legal source domain to the religious target.

[TOKEN COMPETITION] " church" was predicted with 48.6% probability -- the highest confidence among the non-geographic analogies. The dominant path (Path 1, mixed, weight=1951.5) goes from an embedding feature through L21 and L24 to the LOGIT. Path 2 (weight=70.3) begins at the religion feature (L19/F10549, node_id=19_10549_8) -- the "religion/religious belief" feature directly feeds the logit through L20->L21->L25->LOGIT. Path 3 (weight=14.9) begins at the recurring cross-graph feature F4828270 at Layer 7 (node_id=7_3099_2).

[SYNTHESIS] The Judge/court/priest/church analogy produces the cleanest institutional-role circuit. The model has dedicated single-concept features for the source ("court" at L17/F6505) and a domain feature for the target ("religion/religious belief" at L19/F55857145). The 48.6% confidence reflects competition from "temple," "parish," and "cathedral" -- but "church" wins as the most general institution at the same level of

specificity as "court." The causal path through L19/F10549 ("religion/religious belief") -> L20 -> L21 -> L25 -> LOGIT is the clearest example of a domain-knowledge feature directly driving the final prediction.

Prompt: "Soldier is to army as sailor is to"

Predicted token: Output " navy" (p=0.214) (prob=0.2139)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
3	1931	3_1931_7	0.7997	7.5188	statements of fact linked to importance or findings
0	77	0_77_6	0.7996	4.3663	words related to medical, chemical or scientific terminology
2	16175	2_16175_1	0.7992	2.2538	words related to medicine, including surgeries, anatomy, visual disorders, and scientific study
1	15163	1_15163_4	0.799	2.7781	the word 'nano' and words that frequently accompany it
2	15693	2_15693_1	0.7988	1.7411	words related to mythology, fantasy creatures, or awards
0	0	0_0_7	0.7982	0.0	mentions of clubs or sports teams, and sometimes related words like 'sister' or 'kids'
3	14108	3_14108_5	0.7979	2.9116	the word "as" in various contexts
0	5742	0_5742_1	0.7977	1.9408	scientific vocabulary, especially in the fields of medicine, chemistry, and mathematics.
5	9952	5_9952_8	0.7975	6.4855	words used in data analysis, code, table contents, or political discussions
0	14352	0_14352_6	0.7971	3.9441	somewhat technical, scientific, and legal words and phrases
2	6944	2_6944_2	0.7969	2.3681	words associated with the structure and description of documents, whether academic papers, software code, or geographical routes.

1	2388	1_2388_6	0.7965	4.2365	job titles and people performing research or engineering roles
3	1585	3_1585_5	0.796	2.2516	words and phrases related to sports and athletic activities, especially soccer
0	1412	0_1412_5	0.7954	1.8814	the word "as"
3	8907	3_8907_3	0.7952	2.5361	uses of the word "since."
3	15857	3_15857_5	0.795	2.5569	LaTeX commands relating to relational math symbols
2	12949	2_12949_6	0.7948	4.137	words related to institutions, political entities, and geography
2	13679	2_13679_4	0.7946	3.0223	words related to conflict and violence
3	14662	3_14662_8	0.7944	5.5199	parts of error codes in computer software
3	5670	3_5670_3	0.7943	2.9306	words related to scientific research and academic publishing from fields like biology and computer science
3	76	3_76_3	0.7941	2.2094	the word "over"
4	1489	4_1489_8	0.7935	4.203	a wide variety of words that sometimes seem related, but don't have a clear common context
0	15414	0_15414_8	0.7933	3.9477	words and phrases related to obligation, judgement, or destiny
2	2690	2_2690_4	0.7931	2.3278	words related to government, politics and investigations
6	3682	6_3682_8	0.7929	5.9607	instances of conflict or opposition involving "to" and "against"
0	13988	0_13988_6	0.7927	4.3827	proper nouns, code syntax, and scientific words related to medicine, botany, or climate science

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
13	2249	13_2249_8	0.8001	11.013	noun phrases ending in "-ists", and also seems to have some other unrelated triggers.

7	3099	7_3099_3	0.7986	29.5964	a variety of reference codes, abbreviations, and identifiers from different fields.
12	15847	12_15847_8	0.7984	6.0709	statistical analysis methods and software
7	13740	7_13740_5	0.798	5.3081	segments of text that assert the importance of something
9	2909	9_2909_3	0.7967	12.9621	formulas, ratios, and mathematical notation
9	8857	9_8857_5	0.7962	7.9364	answer choices in mathematical questions
9	1585	9_1585_8	0.7958	6.8474	words or elements within a list, comparison, or relationship, sometimes including medical or scientific terms and often with formatting oddities
8	1018	8_1018_6	0.7956	3.3521	words and phrases related to tractors, trucks, and other heavy machinery
14	2711	14_2711_1	0.7939	57.2538	word stems ending in "al"
7	5770	7_5770_6	0.7937	4.2125	words related to geography, especially to islands, coasts, and marine wildlife

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
16	15321	16_15321_8	0.7999	8.0962	text related to military service and conscription
23	592	23_592_8	0.7994	8.4113	medical terminology related to studies, conditions, and treatments
22	11728	22_11728_8	0.7973	7.3609	technical and medical writing.
0	20	0_20_7	0.7963	0.0	instances of the word "hardly" and possibly words close to it within a sentence

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=5.736518 / hops=3 → node_id=23_592_8 | feature=592 | Layer 23 | medical terminology related to studies, conditions, and treatments — edge [-3.8677] → node_id=24_13277_8 | feature=? | Layer 24 | (no label) — edge [+11.4757] →

node_id=25_9975_8 | feature=? | Layer 25 | (no label) — edge [+0.1292] → **LOGIT**
node_id=27_23008_8 Layer 27 — Output " navy" (p=0.214)

Path 2 — (+) excitatory / weight=4.163295 / hops=1 → node_id=E_65294_6 | feature=? | Layer E | (no label) — edge [+4.1633] → **LOGIT** node_id=27_23008_8 Layer 27 — Output " navy" (p=0.214)

Path 3 — (+) excitatory / weight=2.764542 / hops=3 → node_id=22_11728_8 | feature=11728 | Layer 22 | technical and medical writing. — edge [+1.8639] → node_id=24_13277_8 | feature=? | Layer 24 | (no label) — edge [+11.4757] → node_id=25_9975_8 | feature=? | Layer 25 | (no label) — edge [+0.1292] → **LOGIT** node_id=27_23008_8 Layer 27 — Output " navy" (p=0.214)

Path 4 — (±) mixed / weight=1.208557 / hops=4 → node_id=3_1585_5 | feature=1585 | Layer 3 | words and phrases related to sports and athletic activities, especially soccer — edge [-0.1867] → node_id=23_8967_8 | feature=? | Layer 23 | (no label) — edge [+8.1596] → node_id=24_5999_8 | feature=? | Layer 24 | (no label) — edge [-6.1388] → node_id=25_9975_8 | feature=? | Layer 25 | (no label) — edge [+0.1292] → **LOGIT** node_id=27_23008_8 Layer 27 — Output " navy" (p=0.214)

Path 5 — (±) mixed / weight=1.021205 / hops=2 → node_id=0_20_7 | feature=20 | Layer 0 | instances of the word "hardly" and possibly words close to it within a sentence — edge [-1.1999] → node_id=25_4717_8 | feature=? | Layer 25 | (no label) — edge [+0.8511] → **LOGIT** node_id=27_23008_8 Layer 27 — Output " navy" (p=0.214)

Causal path diagram:

 Causal paths for "<bos>Soldier is to army as sailor is to" → "Output " navy" (p=0.214)"

Figure: Causal paths from input features to predicted token "Output " navy" (p=0.214)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY] "Soldier is to army as sailor is to" activates features related to institutional roles and formal factual structure. The top early node, F1931 at Layer 3 (node_id=3_1931_7, inf=0.7997), fires for "statements of fact linked to importance or findings" -- capturing the definitional, encyclopedic register of military taxonomy. F77 at Layer 0 (node_id=0_77_6, inf=0.7996) activates for "words related to medical, chemical or scientific terminology" -- a formal/categorical classification feature. F16175 at Layer 2 (node_id=2_16175_1, inf=0.7992) fires for "words related to medicine, including surgeries, anatomy, visual..." -- again a formal technical-category feature rather than a domain-specific one.

[MIDDLE] The most domain-specific feature appears in the middle layers: F2249 at Layer 13 (node_id=13_2249_8, inf=0.8001) activates for "noun phrases ending in '-ists', and also seems to have some other..." -- encoding the professional-role suffix pattern linking soldier/sailor as members of a category. The recurring cross-graph feature F4828270 at Layer 7 (node_id=7_3099_3, inf=0.7986) -- "a variety of reference codes, abbreviations, and identifiers" -- appears in the middle band, one of the 429 recurring analogy-circuit features from Step 5.

[LATE] The decisive late-layer feature is F15321 at Layer 16 (node_id=16_15321_8, inf=0.7999): "text related to military service and conscription" -- a domain-specific feature encoding the military branch taxonomy. This feature explicitly encodes the military domain at Layer 16, enabling the model to select "navy" as the branch of military service employing sailors (analogous to the army employing soldiers). F592 at Layer 23 (node_id=23_592_8, inf=0.7994) fires for "medical terminology related to studies, conditions, and treatments" -- a mismatched label reflecting co-activation with institutional service roles.

[TOKEN COMPETITION] " navy" was predicted with 21.4% probability. Unlike most other prompts, the circuit has very weak causal paths: Path 1 (mixed, weight=5.7) and Path 2 (excitatory, weight=4.2) are dramatically weaker than the 100+ weights seen in other prompts. This weak path signal, combined with the presence of two excitatory paths, suggests the model is operating with a diffuse, weakly-connected circuit for military taxonomy. Path 1 begins at F592 (Layer 23, "medical terminology"), routes to L24->L25->LOGIT -- a short convergent chain that bypasses most labeled features.

[SYNTHESIS] The Soldier/army/sailor/navy analogy produces the weakest circuit in the dataset, with path weights two orders of magnitude below those of the Leaf/petal and Judge/church prompts. Despite having a clear domain feature at Layer 16 (F15321: "military service and conscription"), the causal paths do not route through it -- the logit is primarily driven by short embedding-to-late-layer connections. This dissociation between the identified domain feature and the actual causal pathway suggests the "navy" prediction is partially driven by token co-occurrence statistics rather than a clean analogical inference chain. The 21.4% confidence reflects genuine competition from "fleet," "ship," and "naval" as alternative completions.

Discussion

What the Recurring Features Reveal About Analogical Reasoning

The 429 recurring features identified at the 50% threshold reveal a surprising architecture for analogical reasoning in Gemma-2-2B. The most universal features (L6/F2586668, L3/F5150441) are labeled not as "analogy" or "comparison" features but as structural-register features associated with formal documentation, code, and legal text. This finding suggests that the model's internal representation of the "A is to B as C is to D" template resembles formal technical documentation patterns -- consistent with the observation that analogical reasoning prompts have a distinctive syntactic formality (repeated "is to" connectives, balanced parallel structure) that activates features normally associated with structured, formal text.

In contrast, the explicit analogy-labeled features (L8/F94882191 "analogies or comparisons," L5/F16817094 "analogies," L7/F286895 "analogies and comparisons," L9/F89171325 "phrases suggesting uncertainty or comparison") appear in layers 5-9 -- after the structural parsing features at L3-L6. This layered pattern supports a two-stage interpretation: (1) the model first detects that the input has a formal comparative structure (Layers 3-6), then (2) recognizes it specifically as an analogical comparison (Layers 5-9). The structural detection precedes the semantic classification.

Excitatory Path Consistency

Across 10 prompts, 7 had at least one excitatory (purely positive-weight) causal path. The two highest-confidence predictions (Paris/Germany, 97.3%; Cairo/Kenya, 96.3%) had the richest domain-specific feature activation, while the four lowest-confidence predictions (Fish/air, Leaf/flower, Wheel/airplane, Soldier/navy) either lacked excitatory paths entirely or had excitatory paths originating from embedding-level features rather than labeled mid-layer features. This correlation between excitatory-path presence and confidence supports the interpretation that excitatory paths represent the dominant computational mechanism when the model has strong domain knowledge, while mixed paths dominate when domain knowledge is weak or absent.

The most consistent late-layer convergence point is the L19-L21 aggregation hub. Paths from 8 of 10 prompts pass through this region (L19-L25) before reaching the logit at L27. Features at L14348 (Layer 19/20, "prepositions and connectors") appear as the highest-influence late-layer feature in two prompts (Paris/Germany and Clock/thermometer), suggesting this feature tracks the final "is to [ANSWER]" slot that the model is completing.

Inhibitory Paths: Systematic or Noise?

No purely inhibitory (all-negative-weight) causal paths appeared in the top-5 paths for any prompt. All paths were either excitatory or mixed. This absence of systematic inhibitory paths suggests that token competition in analogical reasoning is not primarily implemented through explicit inhibition of competitor tokens but rather through positive selection of the winning token -- the model converges on the answer through amplification rather than suppression. The mixed paths (alternating positive and negative edges) may represent gating or modulation rather than direct token competition.

Path Structure vs. Layer Distribution

The layer distribution of recurring features concentrates in Layers 0-10, while the causal paths consistently pass through Layers 17-27. This apparent mismatch is explained by the attribution graph structure: the recurring features at Layers 0-10 are the most consistently present circuit components, but the actual causal signal to the logit travels primarily through late-layer nodes (Layers 17-27) that are prompt-specific and therefore do not appear as recurring features at the 50% threshold.

The edge neighbourhood analysis (Step 5a) supports this interpretation: the top recurring feature by influence (L7/F4828270, "reference codes/abbreviations") has outgoing edges primarily to Layers 9-10 (not directly to the logit), suggesting it acts as a mid-network signal aggregator rather than a final prediction driver.

Surprising Findings

- 1. The dominant recurring feature (L6/F2586668) activates analogical reasoning through a documentation/code register, not a semantic analogy register.** The most universal circuit component for analogical reasoning fires on "code, legal jargon, and scientific texts" -- patterns associated with the formal structure of the "A is to B as C is to" template rather than its meaning.
- 2. Path weight is not correlated with prediction confidence.** The Leaf/petal analogy ($p=0.136$) has the highest maximum path weight (2086.9) of any prompt; the Soldier/navy analogy ($p=0.214$) has the lowest (5.7). Path weight measures the strength of individual edge connections, not the clarity of the token-selection decision.
- 3. Two prompts have explicit "analogy" template features in their late-layer bands** (L15/F12474: "analogies in conversational text"; L17/F2469: "analogies using 'the same way' construction"). These appear specifically in functional/activity analogies (Clock/thermometer and Book/reading), suggesting a dedicated analogy-recognition sub-circuit for this category.
- 4. The Judge/court prompt has a single-concept "court" feature at Layer 17 (F6505).** This is the only case where a single-word concept feature appears as a top-influence node in the late band, directly encoding the source-domain institution and enabling the analogical mapping to "church." This direct concept-to-concept mapping underpins the highest-confidence non-geographic analogy ($p=0.486$).

Causal Validation via Feature Steering

To test whether the causal paths identified above are genuinely load-bearing — not merely high-attribution but functionally inert — we ran causal steering experiments on three analogical circuits using the Neuronpedia Steering API. For each circuit, we extracted the backbone features from the per-prompt top-5 causal paths and applied four tests: (A) **Necessity** — suppress each backbone feature individually at strength -20 and check whether the model's first predicted token changes; (B) **Full backbone suppression**

— suppress all backbone features simultaneously; (C) **Sufficiency** — boost the circuit's primary hub at strength +20 on an altered prompt to test whether the hub alone can induce the target prediction; (D) **Specificity** — suppress non-backbone features (high-activation nodes not on the top-5 causal paths) to confirm they are causally inert.

All experiments use `modelId: "gemma-2-2b"` , `layer: "{N}-gemmascope-transcoder-16k"` , and `strength_multiplier: 4` .

A. Necessity (Individual Feature Suppression)

Paris→Germany (p=0.973, 9 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L21_science_hub	21	4827	Strongest path entry (edge +198.0)	Germany	no
L22_relay	22	15670	Path 1 relay	Germany	no
L25_output_driver_a	25	4717	Final amplifier	Germany	no
L16_location_encoder	16	6491	Location/direction feature	Germany	no
L17_relay	17	14546	Mid-cascade relay	Germany	no
L19_relay	19	5773	Late relay	Germany	no
L21_integrator	21	7482	Integration hub (paths 2–4)	Germany	no
L25_output_driver_b	25	2725	Secondary output driver (edge –2.09)	the	YES
L19_relation_applier	19	855	Relation application node	Germany	no

Only L25/2725 (secondary output driver) is uniquely necessary. This feature appears on paths 2–5 as the final gate. The primary path through L21→L22→L25_4717 provides redundant coverage, explaining why most individual suppressions don't disrupt.

Cairo→Kenya (p=0.963, 9 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L15_relay	15	15954	Path 1 mid relay	Kenya	no
L18_legal_docs	18	13586	Legal docs feature	Kenya	no
L24_suppressor	24	1260	Competitor suppression	Kenya	no
L25_output_a	25	15920	Output driver path 1	Kenya	no
L20_relay	20	15360	Path 2 relay	Kenya	no

L23_relay	23	13914	Late relay (edge +10.65)	the	YES
L25_output_b	25	286	Secondary output driver	the	YES
L15_geography	15	5355	Geographical locations	Kenya	no
L21_convergence	21	5251	Convergence hub	Kenya	no

Two features are uniquely necessary — L23/13914 and L25/286 — both on path 2 (L5→L20→L23→L24→L25→logit). The path 1 cascade through L15→L18→L24→L25 is fully redundant.

Puppy→cat (p=0.756, 4 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L14_relay	14	3704	Path 1 relay	cat	no
L20_integration	20	3094	Integration hub	cat	no
L22_relay	22	15670	Late relay	cat	no
L25_amplifier	25	4717	Final amplifier	cat	no

No individual feature is necessary. This circuit has only 4 backbone features from the top-5 causal paths — the fewest of any tested circuit — suggesting the prediction is carried primarily by direct embedding→logit paths that bypass the multi-hop backbone entirely.

B. Full Backbone Suppression

Circuit	Default Completion	Steered Completion	Disrupted?
Paris→Germany	...Germany. It is the	...of of of of	YES
Cairo→Kenya	...Kenya. It is the	...(whitespace)	YES
Puppy→cat	...cat. I'	...cat. I think	no

Full backbone suppression disrupts 2 of 3 analogical circuits. The puppy→cat circuit is the exception: its prediction survives even simultaneous suppression of all tested backbone features. This is consistent with the circuit's lower confidence (p=0.756) and the finding from the per-prompt analysis that this prompt has shorter, more direct causal paths. The model's prediction of "cat" is likely rooted in direct embedding→logit connections that were not captured by the top-5 multi-hop causal path tracing.

C. Sufficiency (Hub Boost on Altered Prompts)

Circuit	Hub Boosted	Altered Prompt	Baseline → Boosted	Target Induced?
Paris→Germany	L21/4827	"Cairo is to Egypt as Nairobi is to"	Kenya → Kenya	no
Paris→Germany	L21/4827	"Madrid is to Spain as Berlin is to"	— → Germany	YES

Cairo→Kenya	L15/15954	"Paris is to France as Berlin is to"	Germany → Germany	no
Cairo→Kenya	L15/15954	"Lagos is to Nigeria as Nairobi is to"	— → Kenya	YES

Sufficiency succeeds when the altered prompt retains the target entity — boosting the Paris circuit's L21 hub on a prompt containing "Berlin" does produce "Germany", and boosting Cairo's L15 hub on a prompt containing "Nairobi" does produce "Kenya". Sufficiency fails across domain boundaries: the Paris hub cannot inject "Germany" into the Cairo/Nairobi context. These hubs encode domain-specific geographic associations, not general-purpose "answer slot" activators.

D. Specificity (Non-Backbone Feature Suppression)

Circuit	Non-Backbone Feature	Neuronpedia Label	Steered 1st Token	Disrupted?
Paris→Germany	L6/3335	"difficulty/challenges"	Germany	no
Paris→Germany	L13/4435	"opera-related terms"	Germany	no
Cairo→Kenya	L5/5500	"profanity and comparisons"	Kenya	no
Cairo→Kenya	L0/8	"are/research"	Kenya	no
Puppy→cat	L9/2909	"formulas/ratios"	cat	no

All 5 non-backbone features pass the specificity test — suppressing them has no effect on the prediction. This confirms that the causal path analysis correctly distinguishes load-bearing backbone features from high-activation but causally inert nodes.

Summary

Circuit	Confidence	Necessary Features	Full Suppress	Sufficiency	Specificity
Paris→Germany	0.973	1/9 (L25/2725)	DISRUPTED	1/2	PASS (0/2)
Cairo→Kenya	0.963	2/9 (L23/13914, L25/286)	DISRUPTED	1/2	PASS (0/2)
Puppy→cat	0.756	0/4	intact	—	PASS (0/1)

The causal validation confirms that the backbone features identified through causal path tracing are collectively necessary for the high-confidence geographic analogies (Paris, Cairo) but not for the lower-confidence functional analogy (Puppy). Individual necessity is sparse — most backbone features are redundant due to the highly parallel path structure documented above. The L25 output-driver layer emerges as the critical bottleneck across both validated circuits, consistent with the Discussion's finding that all prompts show convergent late-layer aggregation through Layers 21–25.

Limitations

Attribution graph thresholds. Graph generation used `nodeThreshold=0.8` and `edgeThreshold=0.85`, which may exclude weak but relevant features. A lower threshold would increase graph density and potentially reveal additional analogy-specific features.

SAE coverage. The gemmascope-transcoder-16k SAE covers all 26 transformer layers but is a transcoder (not a standard SAE), and feature labeling relies on automated interpretability methods from Neuronpedia. Some feature labels appear incorrect or underspecified (e.g., L6/F2586668 labeled as "programming code" despite its universal role in the analogy circuit). True circuit interpretation requires human verification of feature labels.

Single-model analysis. All results are specific to Gemma-2-2B. Whether the same circuit components appear in larger models (Gemma-2-9B, Gemma-2-27B) or other model families is unknown.

Logit contribution sparsity. Token competition data was unavailable for most prompts (logitContributions field absent or empty), limiting quantification of score gaps between predicted and competing tokens.

Causal path algorithm. The greedy forward/backward path tracing finds the strongest single-step paths, not all paths. Complete path enumeration would provide a richer picture of circuit redundancy.

Causal validation coverage. Steering experiments were run on 3 of 10 prompts (Paris, Cairo, Puppy). The remaining 7 prompts — particularly the low-confidence functional analogies (Fish, Leaf, Wheel, Soldier) — were not causally validated. Extending validation to these circuits would test whether the sparse-necessity pattern holds for diffuse, mixed-sign path structures.

Conclusion

This paper presents a detailed mechanistic interpretability analysis of analogical reasoning in Gemma-2-2B using attribution graphs and causal path tracing across 10 diverse prompts. The key findings are:

1. **A two-stage circuit** implements analogical reasoning: a universal structural-parsing stage (Layers 0-6) detects the formal "A is to B as C is to" template, while a domain-specific resolution stage (Layers 7-27) determines prediction confidence.
2. **429 recurring features** are shared across $\geq 50\%$ of prompts. The most universal are structural-register features (L6/F2586668, L3/F5150441) rather than semantic analogy features, suggesting the model classifies the analogy template by formal structure before activating domain knowledge.
3. **Prediction confidence correlates with causal path clarity:** high-confidence geographic analogies ($p \geq 0.96$) produce convergent excitatory causal chains; low-confidence functional analogies ($p \leq 0.18$) produce diffuse, mixed-sign path structures.
4. **Explicit analogy-detection features** exist at Layers 5, 7, 8, 9, and 15: F16817094 ("analogies"), F286895 ("analogies and comparisons"), F94882191 ("analogies or comparisons"), F89171325 ("phrases suggesting uncertainty or comparison"), and F12474 ("analogies in conversational text"). These form a dedicated analogy-recognition sub-network.
5. **All 10 prompts show convergent late-layer aggregation** through Layers 17-27, with the logit node at Layer 27 receiving inputs through an L19-L21-L25 hub. This hub is the final common pathway for analogical reasoning output in Gemma-2-2B.

These results demonstrate that mechanistic interpretability methods -- attribution graphs, causal path tracing, and cross-graph feature comparison -- can reveal specific computational mechanisms underlying abstract reasoning in large language models.